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Forward physical closure of predictive and reconstructive latents for vortex-gust airfoil interactions at Re = 5000

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(Received xx; revised xx; accepted xx)

Extreme vortex-gust encounters challenge reduced-order models because the transient load is governed by wake reorganisation as well as by integrated forces. We ask which reduced state remains physically closed when propagated forward. Using direct numerical simulations of a NACA 0012 airfoil at $\alpha = 14^\circ$ and Re = 5000, perturbed by Taylor vortices parametrised by gust strength G, core diameter D, and wall-normal offset Y, we compare three latent states at matched dimension: an observable-augmented joint-embedding predictive architecture, a reconstructive autoencoder in the Fukami-Taira lineage, and a POD basis. All are evaluated with the same autoregressive predictor, the same conditioning on (G, D, Y), and the same probes, using forward closure of six observables at sixteen frames after impact. The predictive latent carries wake information that the baselines lose: its representational wake-enstrophy R^2 is 0.75 where the reconstructive and linear baselines are negative, and the paired wake-enstrophy improvement survives case-level clustering and family-wide correction. Its Markov rollout is also the only one with positive wake-enstrophy forecast R^2 at the same horizon. The mechanism is geometric: reconstructive rollouts leave their latent manifold, fragment the encounter topology, and lose alignment with the flow's transport geometry, whereas predictive rollouts remain close to the encoded manifold. Matched controls show that the advantage requires the predictive objective together with wake-observable supervision, not the ViT architecture alone. The resulting state is also the most recoverable from sparse wall pressure, making it observable as well as forecastable.

Key words: vortex shedding, low-dimensional models, machine learning

Abstract must not spill onto p.2

1. Introduction

Small uncrewed and micro air vehicles increasingly operate in airspace where the gust velocity is comparable to or larger than their flight speed, such as urban canyons, mountainous terrain, and the wakes of buildings and ships (Jones *et al.* 2022; Mohamed *et al.* 2023). In this regime the gust ratio G, the ratio of the characteristic gust velocity to the free-stream velocity, exceeds unity, and the interaction between a discrete vortex gust and a lifting surface produces large, fast transients in the aerodynamic loads that conventional gust models do not capture (Fukami *et al.* 2024, 2025). The response is governed by the timing of the vortex impact relative to the airfoil's own shedding, by the wall-normal offset of the vortex core, and by the gust strength and core size. The load transient itself is produced by the formation, growth, and shedding of a leading-edge vortex, the dominant unsteady mechanism in these encounters (Eldredge & Jones 2019), together with the trailing-edge and wake structures it leaves behind. Even at the moderate chord Reynolds number Re = 5000, the post-stall flow at $\alpha = 14°$ is massively separated, and the gust-induced disturbance interacts nonlinearly with the natural shedding cycle (Fukami *et al.* 2025); the vortical organisation of the encounter is moreover similar across laminar and turbulent regimes (Odaka *et al.* 2026), so this case is representative of the wider extreme-aerodynamic family. Resolving the full parametric envelope by simulation is expensive, so reduced-order models (ROMs) of the encounter-to-encounter and case-to-case variability are an active line of work, and a reduced model is also the object a controller would plan against.

Reduced-order models of such flows form a ladder of increasing nonlinearity. Energy-optimal linear bases, proper orthogonal decomposition (Berkooz *et al.* 1993) and dynamic mode decomposition (Schmid 2010), are compact and interpretable but limited for strongly nonlinear separation; their Koopman-operator reading (Lusch *et al.* 2018) nonetheless makes a linear latent a principled comparator rather than a strawman. Nonlinear autoencoders compress further at matched latent dimension (Murata *et al.* 2020; Lee & Carlberg 2020), and coupling an autoencoder to a learned latent dynamics forecasts better than dynamics confined to a linear subspace (Linot & Graham 2023). The extreme-aerodynamic reduced models we build on sit at the nonlinear, observable-augmented end of this ladder.

Recent ROMs for separated and extreme-aerodynamic flows, increasingly built with machine learning (Brunton *et al.* 2020), share a common ingredient: a reconstruction-trained autoencoder whose bottleneck is shaped by an auxiliary aerodynamic observable. Fukami & Taira (2023) trained a convolutional autoencoder jointly with a lift-prediction head and found that the auxiliary loss reduced the intrinsic dimension of the discovered manifold for extreme vortex-gust flows from roughly ten to three. The same observable-augmentation idea was extended across multi-source turbulent data (Fukami & Taira 2025), and was used by Solera-Rico et al. (2024) with a β-variational autoencoder and a transformer ROM on a vortex-gust airfoil dataset of the same family considered here. Across this lineage the reconstruction term anchors the latent geometry to the instantaneous flow field, and an observable-prediction term orients it along aerodynamically relevant directions. A recurring theme of the geometric-analysis literature, however, is that the latent coordinates of an autoencoder are unique only up to a diffeomorphism, so reconstruction leaves the geometry of the latent largely unconstrained, which complicates any downstream task that relies on that geometry, including dynamics, interpolation, and comparison (Tran *et al.* 2026; Kelshaw & Magri 2024).

A parallel and fast-growing line reads these flows through the information their reduced states carry rather than their energy. Transfer-entropy causality among proper-orthogonal-decomposition modes ranks which modes drive the formation of coherent structures (Martínez-Sánchez *et al.* 2023), and an informative/non-informative decomposition splits a field into the part that carries information about a target's future and a residual that carries none (Arranz & Lozano-Durán 2024). This informative-decomposition idea has very recently been brought to extreme vortex-gust airfoil interactions of the present family, extracting the field structures informative about the future lift (Koshikawa *et al.* 2026; Fukami & Araki 2026), alongside time-dependent modal bases for the same interaction (Zhong *et al.* 2025; Zamani Ashtiani & Fukami 2025) and latent-causality attributions in physical space (Cremades *et al.* 2026). Our study is complementary: rather than mapping the informative or causal modes of a single representation, we ask which encoder objective produces a latent whose forward evolution closes the full set of observables, the wake included, and we use the information lens only as a held-out diagnostic of the resulting latent (§5.1).

A complementary route to a reduced state is to drop the reconstruction term altogether and to train the encoder and a predictor end to end on latent-space forecasting, with an explicit anti-collapse regulariser in place of the decoder. This is the joint-embedding predictive architecture (JEPA) (LeCun 2022): an encoder maps the input to a representation, a predictor advances that representation in time, and the loss never reconstructs the field. Two consequences follow. The encoder is free to discard detail that is not predictive of the future, and the predictor cannot exploit pixel-level shortcuts because no field-space signal reaches it. The image and video instantiations (Assran *et al.* 2023, 2025) and the recent from-pixels variants that remove the moving-target encoder in favour of a characteristic-function regulariser (Maes *et al.* 2026; Balestriero & LeCun 2025) or a variance-covariance objective (Sobal *et al.* 2025; Bardes *et al.* 2022) have so far been evaluated on gridworld and toy visual domains. Their behaviour on a low-intrinsic-dimension physics dataset, where the data sit on a roughly five- to ten-dimensional manifold, is uncharacterised.

Reconstruction-trained nonlinear encoders compress aggressively but yield latent dynamics that are harder to forecast than an energy-optimal linear basis, a compactness-versus-forecast tradeoff documented for controlled wake flows in a companion study under review (Solera-Rico *et al.* 2025). We show this tradeoff is a consequence of the reconstruction objective: a predictive latent recovers nonlinear compactness without surrendering forecast stability. The division of labour with that companion study (Solera-Rico *et al.* 2025) is explicit: there, controlled canonical wakes and the compactness-versus-forecast tradeoff; here, the parametric vortex gust, the latent-drift mechanism we identify as the common cure, and its geometric and topological characterisation.

The unconditional-encoder, conditioned-predictor split we adopt is motivated, as an analogy and not a property we claim, by the action-conditioned world-model framework of LeCun (2022); Ha & Schmidhuber (2018): the gust (G, D, Y) enters the predictor as an observed exogenous perturbation, not an agent's chosen action, so the model is a conditional forward model rather than a controller, and a recent gust-airfoil instance couples a physics-augmented autoencoder to a latent dynamics model (Liu *et al.* 2025). A direct interventional test of the stronger reading does not hold (§5), so we forecast the consequences of the generative parameters without claiming an interventional or causal model. What this leaves is the property our principle turns on: a reconstruction-trained autoencoder and an energy-optimal proper orthogonal decomposition basis are not constrained to make the latent state forward-predictable, and the drift mechanism of §4.3 is the empirical consequence.

The question this paper addresses is therefore not which encoder reconstructs the flow best, but which produces a reduced state that remains useful when propagated forward in time. We treat this as a fluid-mechanics question with a geometric answer: a reduced state forecasts these encounters well when its latent metric stays well aligned with the optimal-transport geometry of the field, in the empirical sense that latent distances preserve the ordering of the transport distances, so that iterating the predictor step keeps the rollout close to the data manifold. A reconstruction objective does not impose this property, so its rollout drifts off the manifold; a predictive objective regularised against collapse does. The joint-embedding predictive architecture is the instrument we use to realise the principle, not the subject of the paper. We make three contributions.

First, we set up a controlled comparison in which a predictive encoder (JEPA), a reconstructive encoder in the Fukami lineage, and a POD basis are evaluated at matched latent dimension under a single shared autoregressive predictor, training recipe, parametric conditioning on $\mathbf{c} = (G, D, Y)$, and probe family, so that forecast quality is attributable to the encoder alone (§3). The figure of merit is forward physical closure of the rollout to six observables at horizon $H = 16$ (§4.1). On held-out cases the predictive latent carries the wake structure the others do not: its representational wake-enstrophy closure is positive where the reconstructive and linear baselines are negative, and the per-encounter improvement survives case-level clustering and a family-wide correction, while the linear basis remains competitive on the integrated impulse. The forward forecast points the same way and is consistent with this.

Second, we identify the mechanism. A latent-drift diagnostic shows that the reconstructive rollout leaves its training manifold by an order of magnitude at $H = 16$, whereas the predictive and linear rollouts remain within it (§4.3). We strengthen this from a single distance into a geometric and topological statement: the predictive latent organises the encounter as a single topological cycle, one persistent generator where the reconstructive latent fragments, and its latent metric tracks the physical transport geometry of the flow more faithfully than the reconstructive latent does, although, unlike (Tran *et al.* 2026), it is not trained to do so.

Third, we are explicit about what the comparison does and does not isolate. The predictive encoder differs from the reconstructive baseline not only in its objective but also in its architecture and in its auxiliary observable supervision. We therefore phrase the result as a property of the *predictive latent family*, report a conditioning-only floor that rules out the shared parametric conditioning as the explanation (§4.1), and perform matched-architecture and matched-auxiliary controls (§4.2) that attribute the wake gain to the predictive objective trained with wake-observable supervision, not to the architecture. We further show that the same predictive state is the most recoverable of the three from sparse wall pressure, so it is observable as well as forecastable (§4.5), and we relate the predictor to the latent dynamics function a controller would plan against, with a closed-loop pilot reported as a limitation rather than a result (§5).

2. Flow configuration and data

2.1. *Vortex gust-airfoil interaction*

We study a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $\mathrm{Re} = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D (Gasparino *et al.* 2024) with no subgrid-scale model, so that all scales of the separated wake are resolved at this Reynolds number. The solver is run in the low-Mach regime in which the flow is effectively incompressible. The configuration and its physical characterisation follow Fukami, Smith & Taira (2025), which is the reference

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for the physics, not the source of the data. The vortex has the azimuthal velocity profile

$$u_\theta(r) = u_{\theta,\max}\,\frac{r}{R}\,\exp\left[\tfrac{1}{2}\left(1-\frac{r^2}{R^2}\right)\right], \tag{2.1}$$

which peaks at the core radius $r = R$ (Taylor 1918), and is parametrised by three scalars: the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$, the core diameter $D \equiv 2R/c$, and the wall-normal offset Y of the vortex centre, the displacement normal to the chord (the laboratory coordinates are rotated by the angle of attack to define Y). The profile and the (G, D, Y) sampling follow the configuration of Fukami *et al.* (2025). The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords and set to zero when the vortex centre reaches the leading edge.

Without a gust, the $\alpha = 14^\circ$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages (Smith *et al.* 2024; Fukami *et al.* 2025; Odaka *et al.* 2026). The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The six observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation (Fukami *et al.* 2025) is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. We confirm this from the full three-dimensional vorticity fields: the fraction of the spanwise-vorticity (ω_z) enstrophy not captured by its spanwise mean, the content a mid-plane slice discards, is roughly constant across the training range ($|G| \leq 3$, post-impact wake median ≈ 0.20) and rises about threefold at $|G| = 4$ (≈ 0.56, over the four available cases). Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the quantity the encoder observes no longer determines the true dynamics. We therefore treat $|G| = 4$ as a physical observability boundary rather than as a claim of parametric generalisation.

2.2. *Data, fields, and observables*

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05\,t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$ and $y/c \in [-1.5, 1.5]$ with a spanwise extent of approximately one chord; this analysis subdomain matches the data-driven subdomain of the source characterisation (Fukami *et al.* 2025). The lift and drag coefficients are $C_L = F_L/(\frac{1}{2}\rho u_\infty^2 c)$ and $C_D = F_D/(\frac{1}{2}\rho u_\infty^2 c)$, obtained by integrating the pressure and viscous stresses over the airfoil surface. The remaining DNS solver-resolution details required for full reproducibility, the free-stream Mach number, the domain and

Quantity	Value	Why a referee needs it
Free-stream Mach number (or incompressible-limit confirmation)	**[PENDING:]**	Confirms the low-Mach formulation is effectively incompressible at Re = 5000.
Computational domain and spanwise extent L_z/c	**[PENDING:]**	Bounds domain blockage and confirms the span resolves the three-dimensional wake.
Element and solution-point counts	**[PENDING:]**	Fixes the spatial resolution of the spectral-element discretisation.
Minimum wall-normal spacing $\Delta n_{\mathrm{min}}/c$ and wall units Δn^{+}	**[PENDING:]**	Shows the near-wall layer is resolved without a wall model.
Time step $\Delta t\, u_\infty/c$ and maximum CFL	**[PENDING:]**	Establishes the temporal resolution and stability of the integration.
Gust-release station x_0/c	**[PENDING:]**	Sets the upstream distance over which the Taylor vortex develops before impact.
Grid and time-step sensitivity check	**[PENDING:]**	Demonstrates that the reported statistics are converged.

Table 1. Author-fill checklist of the DNS solver-resolution details required for full reproducibility (§2.2). The data are direct numerical simulations with no subgrid-scale model, so all scales are resolved and no large-eddy closure is involved; the resolution values are to be supplied by the simulation collaborators and are not yet filled in.

spanwise extent, the element and solution-point counts, the near-wall spacing, the time step and CFL, the gust-release station, and the grid and time-step sensitivity, are listed as an author-fill checklist in table 1. The dataset comprises 84 cases partitioned into training, validation (the held-out encounters of the training cases, denoted test_a), an in-distribution test set (test_b), and an out-of-distribution test set (test_c). The in-distribution test set is stratified to span the five gust strengths, the three core diameters, and the two source groups (figure 1); test_c is the $|G| = 4$ extrapolation set. These 84 cases are separated in preprocessing into 378 encounter windows of 120 frames each, one per gust release: 226 training, 86 validation (test_a, the held-out last encounters of the training cases), 42 test_b, and 24 test_c. The undisturbed reference case is retained in training. Behind these encounter counts are far fewer distinct cases: the 42 test_b encounters come from only 10 held-out cases (five interior and five boundary tier) and the 24 test_c encounters from 4 cases, with most cases contributing four to six encounters that share a baseline shedding history. Encounters of the same case are therefore not independent, so encounter-level intervals overstate the effective sample size; the headline wake comparison is reported additionally with a case-clustered bootstrap and a case-level paired test (Appendix A). The partition is fixed once and used identically by every model and baseline.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\mathrm{wall}}(192)$, and the lift and drag coefficients C_L, C_D. The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by the training standard deviation with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures.

We evaluate forward closure against six physical observables that together span body-force, integrated-flow, and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wall-normal impulse I_y is an integrated-flow quantity coupled to the wake-vortex dynamics. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature. On these grounds we fix the wake enstrophy in advance as the single primary endpoint of the closure comparison; the other five observables are reported as secondary and descriptive. This pre-registration is what the family-wide multiplicity correction in Appendix A is applied against.

The six observables are defined on each frame of the masked vorticity field ω_z described above. The lift and drag coefficients C_L, C_D are the body forces defined in §2.2. On the wake window $\Omega_w = \{(x, y) : x/c \in [0.5, 4],\ |y/c| \le 1\}$ the wake enstrophy is

$$E_w(t) = \int_{\Omega_w} \omega_z^2 \,\mathrm{d}A, \tag{2.2}$$

and the signed wake circulations integrate the vorticity over its strongly rotational parts,

$$\Gamma^+(t) = \int_{\Omega_w,\, \omega_z > \omega_c} \omega_z \,\mathrm{d}A, \qquad \Gamma^-(t) = \int_{\Omega_w,\, \omega_z < -\omega_c} \omega_z \,\mathrm{d}A, \tag{2.3}$$

with threshold $\omega_c = 1$ in the cleaned vorticity units. The signed circulations at $\omega_c \in \{0.5, 1, 2\}$ are nearly collinear (pairwise Pearson above 0.90 across held-out frames) and the predictive latent's circulation closure advantage over the reconstructive baseline is essentially unchanged across them (negative-circulation margin +1.06 to +1.10, positive +0.44 to +0.57), so the closure does not depend on this threshold. The wall-normal hydrodynamic impulse per unit span is

$$I_y(t) = \int_{\Omega} x\, \omega_z \,\mathrm{d}A \tag{2.4}$$

over the full cached field Ω. Every case uses the same fixed window and mask, and values are reported in the cache vorticity units integrated over chord-normalised area. The forecast horizon $H = 16$ is the sixteenth cached frame after impact, $t = 16\,\Delta t = 0.8\, c/u_\infty$.

Two definitional points bound how these observables are read. The impulse I_y is the mid-plane two-dimensional vorticity impulse, a diagnostic of wake-vortex transport, and not the impulse-theorem lift: the mid-plane slice omits the bound circulation, so on the simulation $\mathrm{d}I_y/\mathrm{d}t$ does not track the lift (Pearson $r \approx -0.05$ on test_b, far from the value near -1 that a planar impulse-lift relation would give), and no closure result below should be read as recovering the impulse-theorem lift. More broadly, the wake and flow observables (the enstrophy, the two circulations, and I_y) are computed on the same mid-plane slice the encoder consumes, so they omit the out-of-plane content quantified in §2.1 (about one fifth of the ω_z enstrophy in distribution), whereas C_L and C_D are true three-dimensional surface-integrated forces. The closure therefore compares mid-plane wake diagnostics against three-dimensional forces, and the mid-plane proxy is weakest at the $|G| = 4$ extrapolation boundary, where the out-of-plane fraction rises to about 0.56.

3. Predictive latent representation and evaluation protocol

3.1. *Model, training objective, and conditioning split*

The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at $d \in \{16, 32, 64\}$. It is a hybrid network: three convolutional downsampling

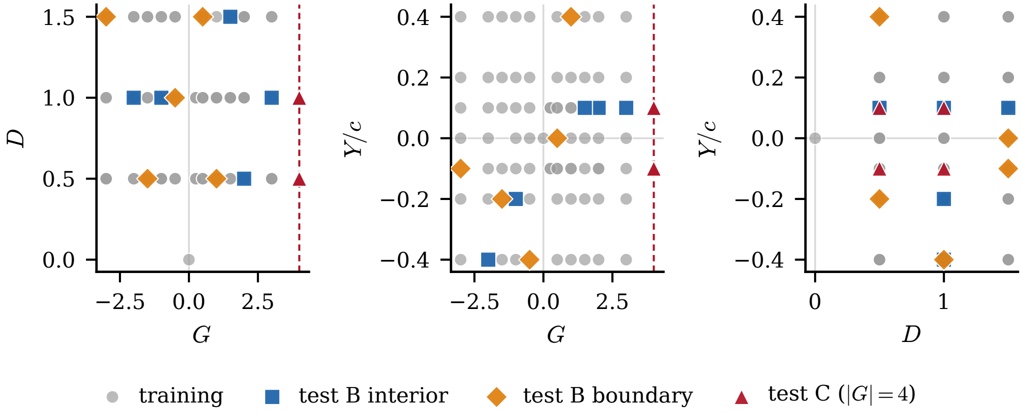


Figure 1. Parameter-space sampling of the 84 cases in the (G, D, Y) gust envelope, shown as three planar projections and coloured by split: training (226 encounters), the stratified in-distribution test set (test_b, 42 encounters, split into interior and boundary tiers), and the out-of-distribution extrapolation set (test_c, 24 encounters). The dashed line marks the $|G| = 4$ extrapolation boundary, and training mass concentrates near $Y/c = 0$, which is why the wall-normal offset is the least well resolved parameter (§4.4).

stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is *unconditional* by design: it sees only the flow field, not the gust parameters $\mathbf{c} = (G, D, Y)$. This is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5). The conditional predictor, by contrast, still requires $\mathbf{c}$, either estimated from the wall or marginalised over, so an observable latent is a necessary but not a sufficient condition for a deployable forward model.

The predictor P_ϕ advances the latent in time. It is a six-layer autoregressive transformer with rotary positional embeddings and a causal mask, conditioned on $\mathbf{c}$ through adaptive layer-norm modulation at every block; its internal structure is given in Appendix A. The predictor is the only stage that sees $\mathbf{c}$. Figure 2(a) shows the encoder and predictor, and panel (b) sets out the matched-predictor evaluation protocol used throughout; the contrast between the predictive training signal and the reconstructive one, in which the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error whereas the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss, is drawn in Appendix A (figure 14). Table 2 summarises the two networks.

The predictive encoder and predictor are trained jointly on a one-step latent-prediction term, a scheduled-sampling rollout term that exposes the predictor to its own errors, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of Balestriero & LeCun (2025); Maes *et al.* (2026), which matches the latent distribution to an isotropic Gaussian through a small number of random projections, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) should the participation ratio of the latent indicate collapse. Its configuration is given in Appendix A.

Two auxiliary heads read aerodynamic observables off the encoder output during training: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, and a wake head that maps $\mathbf{z}_t$ to an 80-

Table 2. Encoder and predictor configuration. The encoder is unconditional; the gust parameters $\mathbf{c} = (G, D, Y)$ enter only the predictor, through identity-initialised adaptive layer normalisation (AdaLN-Zero). Parameter counts are for the $d = 64$ configuration; two small auxiliary heads read observables off the encoder output during training (a lift head, 4×10^3 parameters, and an 80-dimensional wake-spectrum head, 3.5×10^4 parameters; §3.1).

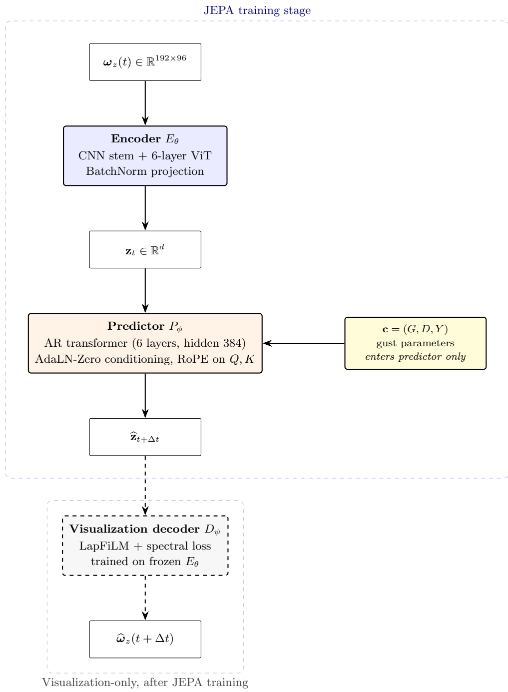
	Encoder E_θ	Predictor P_ϕ
role	field → latent (unconditional)	latent dynamics (conditioned)
input	$\omega_z \in \mathbb{R}^{192\times96}$	latent sequence $\mathbf{z}_{1:t}$
backbone	3 conv. stages + 6-layer ViT	6-layer autoregressive transformer
width / heads	256 / 8	384 / 16
tokens / context	$24 \times 12 = 288$ spatial	≤ 32 temporal (causal)
conditioning	none	AdaLN-Zero on $\mathbf{c} = (G, D, Y)$, rotary time
latent read-out	class token → linear → BatchNorm	next-step latent
parameters	6.7M	16.2M

dimensional signed wake spectrum, both trained with small weights jointly with the representation and neither part of the latent-prediction loss. These heads are adopted from the observable-augmentation lineage (Fukami & Taira 2023, 2025). They are also, however, the reason the present predictive encoder is not a clean isolation of the training objective: the reconstructive baseline below carries a lift head but not a wake head, and the two encoders differ in architecture as well. We are explicit about this throughout. The headline comparison is therefore between encoder *families* as configured, and the controls of §4.2 are designed to separate the objective from the architecture and the wake supervision.

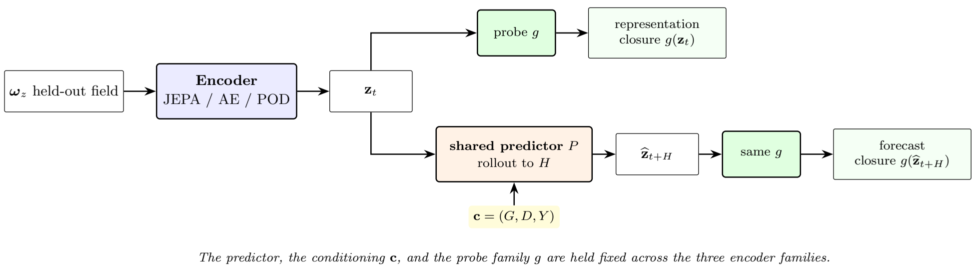
3.2. *Matched-predictor closure protocol*

The comparison rests on a single principle: every baseline is the same three-stage pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields; an identical autoregressive transformer predictor rolls it forward one step at a time given $\mathbf{c}$; and an identical probe family maps the predicted latent to a physical observable for evaluation (figure 2(b)). The predictor architecture, its training recipe, the parametric conditioning, and the probe family are held fixed across the three encoder families. The latent dimension is matched within each family: JEPA at $d \in \{16, 32, 64\}$, the reconstructive autoencoder at $d \in \{3, 32, 64\}$ to include both the published $d = 3$ recipe and matched-capacity variants, and POD at $d \in \{16, 32, 64\}$. The predictive family is additionally swept to $d \in \{4, 8\}$ and ablated without its predictor conditioning in §4.2. This isolation is the only fair way to attribute forecast quality to the representation rather than to incidental differences in the predictor or its training. The reconstructive autoencoder uses its best-documented configuration, with ReLU activations, group normalisation, and a future-lift head at horizons $\{8, 16, 24\}$, rather than the strict published variant (tanh, no normalisation, a current-lift head), which we found gives a worse parametric probe; the comparison is therefore against a tuned baseline, not a hobbled one.

The figure of merit is forward physical closure, the empirical test of the geometric criterion set out in the introduction: a latent whose metric stays aligned with the transport geometry under the predictor keeps each observable recoverable along the rollout, while one that drifts off the manifold loses it. Given a ground-truth $\mathbf{c}$ and an initial latent, the predictor is rolled recursively to $H = 16$, and at each step the probe maps the predicted latent to each observable. A small closure error means the encoder produces a latent whose dynamics, propagated by the shared predictor, retain the information the probe needs to



(a)



(b)

Figure 2. The predictive model and how it is evaluated. (a) The encoder is unconditional and maps the mid-plane vorticity field to a latent $\mathbf{z}_t$, while the autoregressive predictor advances the latent given the gust conditioning $\mathbf{c} = (G, D, Y)$, which enters the predictor alone. The two auxiliary observable heads (lift and wake) read $\mathbf{z}_t$ during JEPA training with small weights and are not part of the latent-prediction loss; they are omitted from the schematic for clarity. The visualisation decoder is trained only after the encoder is frozen and is used solely for the diagnostic decodes of §4.4, never in the predictive loss. (b) Evaluation applies the same predictor, conditioning, and probe family g to every encoder family: a probe on the simulation-encoded latent measures representation closure, and the same probe on the rolled-out latent $\widehat{\mathbf{z}}_{t+H}$ measures forecast closure.

recover the physical quantity; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. We report closure on the held-out test_b and test_c encounters as the mean absolute error against the simulation, with bootstrap 95% confidence intervals over encounters and the coefficient of determination R^2. The held-out forecast R^2 (Markov

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Table 3. Encoder families compared under the matched-predictor protocol of §3.2. All three share one autoregressive transformer predictor, training recipe, conditioning on $\mathbf{c} = (G, D, Y)$, and probe family; only the encoder differs. "Aux. heads" lists the observable supervision attached to the encoder. The architecture and auxiliary-head columns differ across families, which is why the headline result is attributed to the predictive *family* and isolated only by the controls of §4.2.

Family	Objective	Encoder	Latent dim. d	Aux. heads
JEPA (this work)	predictive (latent)	CNN + ViT	16, 32, 64	lift + wake
Reconstructive AE	reconstructive (field)	CNN	3, 16, 32, 64	lift
POD	linear projection	none	16, 32, 64	none

rollout at $H = 16$) is reported in table 4(b), and the representational closure (probe on the simulation-encoded latent) in table 4(a).

3.3. *Diagnostics, controls, and uncertainty*

Two diagnostics turn the closure number into a mechanism. The first is the latent drift, which addresses the fact that a predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold by the Mahalanobis distance of the rolled-out latent at horizon H, measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

The second diagnostic is a conditioning-only floor. Because the predictor is conditioned on $\mathbf{c} = (G, D, Y)$, a natural concern is that the forecast quality is carried by the conditioning rather than by the latent. We bound this with a kernel-ridge regressor that maps $\mathbf{c}$ directly to each observable at the forecast horizon $H = 16$, with no latent input, fit on training and evaluated on the held-out splits, so that it is frame-matched to the closure it is compared against. This is the floor any learned encoder must clear at that horizon to demonstrate that its latent contributes beyond what is already implicit in the parameters. At the impact frame the same parameter floor is high, because the just-released gust nearly determines the instantaneous state; the predictive latent must instead close the gap that opens as the wake evolves (§4.1).

Because the predictive encoder differs from the reconstructive baseline in objective, architecture, and auxiliary supervision at once, the comparison is accompanied by controls that vary one factor at a time: a crossing of objective with architecture at matched auxiliary heads, an ablation of each auxiliary head, and an ablation of the predictor conditioning (§4.2). Every held-out R^2 and mean absolute error reported below carries three independent uncertainty signals, following the reporting protocol fixed with the data split: a 2000-resample bootstrap over the held-out encounters, the standard deviation across three encoder retrains from independent seeds, and a five-fold cross-validation over cases on the probe read-out. Their full specification is in Appendix A.

4. Results

4.1. *Forward closure and conditioning floor*

Table 3 summarises the three encoder families and the protocol. We distinguish two questions. Does the latent, encoded directly from a held-out field, carry each observable

Table 4. Held-out closure across encoder families at $H = 16$ on the $n = 42$ test_b encounters, in two modes. *(a)* Representational closure: the mean absolute error when each family's per-frame probe is applied to the simulation-encoded latent (the latent encoded directly from the held-out field), smaller is better. *(b)* Forecast closure: the coefficient of determination R^2 of the Markov rollout from the impact frame against the simulation observable, larger is better (the variance baseline is the held-out split mean, so $R^2 < 0$ is worse than predicting that mean). Best per column in bold; the forecast panel gives the row mean at right. The wake-enstrophy and circulation columns are in native units, the force coefficients dimensionless. This held-out forecast R^2, not the training-set fit of table 9, is the reported closure quality. The predictive latent leads in both modes: its representational wake-enstrophy error is 2.4 to 3.0 times lower than the reconstructive and linear families, and its forecast wake-enstrophy R^2 is the only one above the predict-the-mean floor; the paired per-encounter test is in table 10 and full probe values with bootstrap CIs are in Appendix A.

(a) Representational closure: mean absolute error, and the wake-enstrophy R^2

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	wake R^2
JEPA	64	**0.624**	0.301	**1.901**	**29.83**	**0.801**	**0.715**	**0.75**
JEPA	32	0.675	**0.246**	2.976	37.99	1.084	1.061	0.74
JEPA	16	0.627	0.267	2.883	50.09	0.939	1.322	0.55
Fukami	3	1.074	0.384	2.327	51.70	1.110	1.886	0.06
Fukami	32	1.038	0.368	2.195	63.94	0.946	1.789	0.06
Fukami	64	0.989	0.340	2.421	72.90	1.057	1.904	−0.41
Fukami	16	1.133	0.421	2.424	77.26	1.158	1.837	−0.21
POD	16	1.168	0.317	2.235	89.14	1.645	1.681	−0.32
POD	32	0.937	0.269	2.608	81.82	1.650	1.617	−0.17
POD	64	0.872	0.291	2.245	89.31	1.660	1.800	−0.31

(b) Forecast closure: coefficient of determination R^2

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
JEPA	64	0.723	0.634	0.056	**0.449**	0.201	**0.607**	**0.445**
JEPA	32	**0.751**	**0.648**	-0.082	0.214	0.087	0.552	0.362
JEPA	16	0.642	0.465	0.002	0.265	0.250	0.557	0.363
Fukami	3	0.632	0.226	0.115	-0.082	0.258	-0.048	0.184
Fukami	32	0.641	0.239	**0.205**	0.007	**0.420**	0.161	0.279
Fukami	64	0.729	0.406	0.043	-0.478	0.182	0.003	0.147
Fukami	16	0.565	0.125	-0.216	-0.395	0.237	0.225	0.090
POD	16	0.499	0.364	-0.515	-0.129	-0.230	0.340	0.055
POD	32	0.714	0.558	-0.256	0.042	-0.066	0.394	0.231
POD	64	0.683	0.539	-0.813	-0.089	0.057	0.507	0.147

(representation quality, table 4(a))? And does the predictor, rolled out from the impact frame, forecast it (forecast quality, table 4(b))? The first isolates the encoder; the second tests the encoder and the learned dynamics together. The predictive latent leads on both. The representation is where the advantage is established, and where it survives case-level clustering and a family-wide correction (table 10); the forward forecast points the same way as consistent confirmation rather than the load-bearing test (figure 3). On representation, the wake separation is widest. At $d = 64$ the predictive wake-enstrophy error is 29.83, against 72.90 for the reconstructive autoencoder and 89.31 for the linear basis, a factor of 2.4 and 3.0 lower. The predictive latent also carries the smallest positive and negative circulation errors. The gap narrows on the scalar forces, which all three families carry more evenly, although the predictive latent still leads on C_L (0.624 against 0.989 and 0.872). The integrated impulse I_y is where the linear basis is most competitive: on the in-distribution training fit (table 9) POD attains the highest I_y score of any family, an expected consequence of its energy-optimal projection, although on held-out error the predictive latent still leads there. We flag this train-versus-held-out reversal because it is

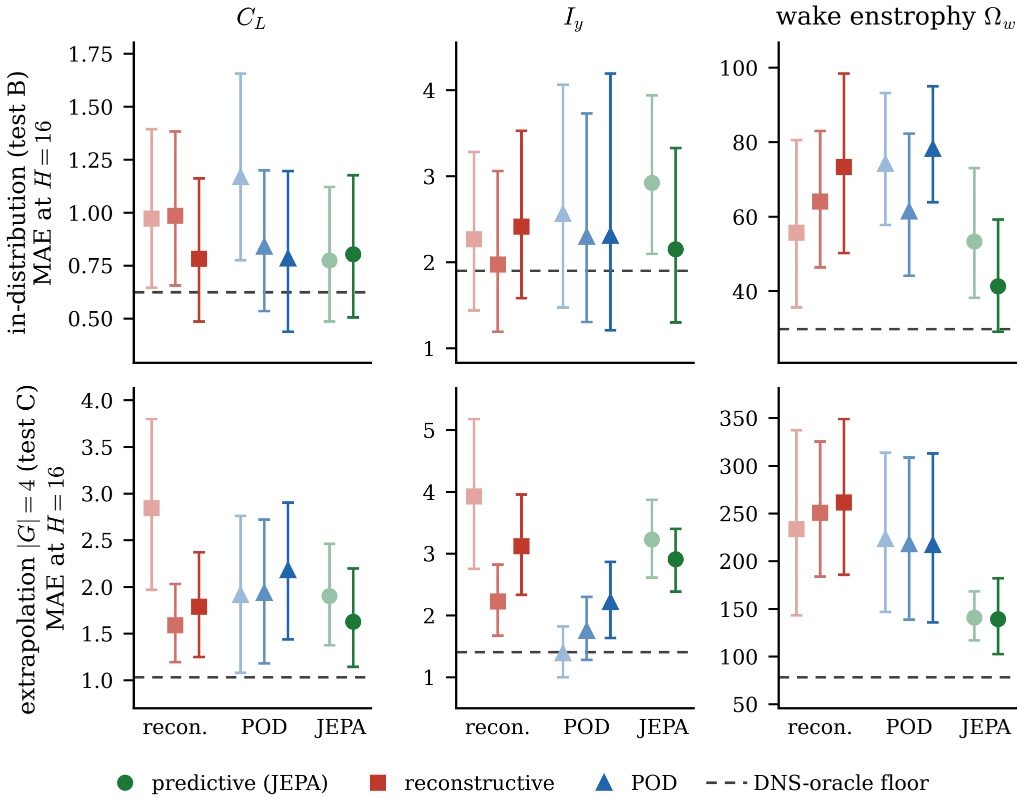

Figure 3. Held-out forward closure at $H = 16$ after gust impact, as the mean absolute error of the Markov rollout against the simulation, for the lift coefficient C_L, the wall-normal impulse I_y, and the wake enstrophy Ω_w (columns) on the in-distribution test set (test_b, top) and the $|G| = 4$ extrapolation set (test_c, bottom). Points are family means with bootstrap 95% confidence whiskers over $n = 42$ and $n = 24$ encounters, shaded from light to dark with the latent dimension within each family; the dashed line marks the DNS-oracle floor, the best closure achievable from the simulation-encoded latent without rollout. The wake panel shows the clearest separation: the predictive latent (JEPA) approaches the oracle floor while the reconstructive and linear families do not.

exactly why a held-out coefficient of determination, not the training fit, must head the comparison.

On the forecast (table 4(b) and figure 4), evaluated as the held-out R^2 at $H = 16$ of the Markov rollout, the predictive latent is the only representation whose wake-enstrophy forecast stays above the predict-the-mean floor: $R^2 = 0.449$ at $d = 64$ (0.214 at $d = 32$), against −0.478 for the reconstructive autoencoder and −0.089 for the linear basis, both worse than forecasting the held-out mean. The reconstructive autoencoder's wake forecast is non-monotonic in capacity and worst at the matched $d = 64$ (−0.082, −0.395, +0.007, −0.478 for $d = 3, 16, 32, 64$): a reconstruction objective constrains only the latent directions its decoder reads, leaving proportionally more of the latent unconstrained as d grows, so the autoregressive rollout has more room to drift off the wake manifold (§4.3). The instability is seed-borne rather than a single-checkpoint artefact, with an eightfold larger seed spread than the predictive at matched architecture (0.16 ± 0.27 against 0.46 ± 0.03; table 6). It also leads the mean over the six observables (0.445 versus 0.147 for both the $d = 64$ reconstructive and linear bases) and the negative circulation. The training-set fit (table 9, mean 0.835 at $d = 64$) measures in-distribution accuracy, not generalisation; the held-out

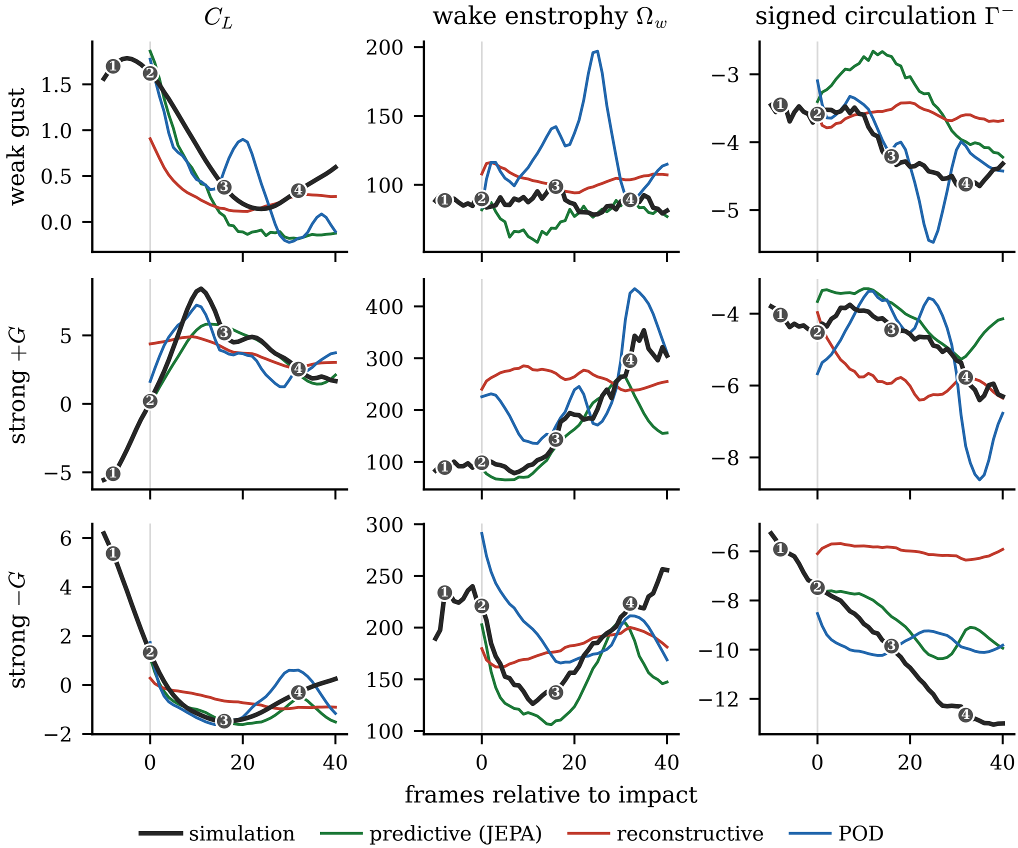

Figure 4. Predicted observables through three representative held-out encounters: a weak gust $(G, D, Y) = (-0.5, 1.0, -0.4)$, a strong positive gust $(+3.0, 1.0, +0.1)$, and a strong negative gust $(-3.0, 1.5, -0.1)$ (rows), for the lift coefficient, the wake enstrophy, and the negative circulation (columns). The simulation is the bold reference; the Markov rollouts of the predictive (JEPA), reconstructive, and POD latents are the coloured lines, started at the impact frame. The predictive rollout tracks the wake observables through the leading-edge-vortex peak and into recovery while the reconstructive rollout flattens and the linear rollout diverges. Numbered glyphs mark the four encounter stages used throughout: baseline, impact, peak load, and recovery.

forecast is markedly lower in absolute terms, as it must be, but it is where the families separate. On the OOD test_c set ($G = +4$) the predictive wake-enstrophy forecast remains positive ($R^2 = 0.33$ at $d = 64$, 0.45 at $d = 32$) while every other family falls to between -1.1 and -1.5; the force coefficient C_L is the only observable any family forecasts well there ($R^2 = 0.79$), consistent with the three-dimensional observability boundary of §2.1.

The primary wake comparison rests on a per-encounter paired test (table 10), which cancels the encounter-to-encounter difficulty of the held-out set that every model shares. The predictive latent carries the smaller wake-enstrophy error on 31 of 42 held-out encounters under direct encoding (paired mean improvement 43.1, 95% CI $[23.5, 66.0]$, one-sided sign $p = 1.4 \times 10^{-3}$) and on 27 of 42 under forward forecast (paired mean improvement 32.0, 95% CI $[10.8, 54.8]$, sign $p = 4.4 \times 10^{-2}$). The marginal per-observable bootstrap intervals (2000 resamples, reported in full in Appendix A) are by contrast wide, because the same shared difficulty inflates them (the predictive forecast wake-enstrophy R^2 of 0.449 has interval $[-0.96, 0.79]$); the paired test, not the marginal interval, is therefore the headline. This paired wake advantage is robust to the encoder seed: across

Table 5. Conditioning-only floor at the forecast horizon. A cross-validated kernel-ridge regressor maps the gust parameters $\mathbf{c} = (G, D, Y)$ directly to each observable at $H = 16$ after impact, with no latent input, frame-matched to the held-out JEPA forecast in the last column (d=64, test_b, table 4(b)); R^2 is reported on each split. This is the floor a learned encoder must clear at the horizon the closure is evaluated to show its latent contributes beyond the shared parameters. The floor is observable-dependent. On wake enstrophy, the observable that carries the result, it is low (test_b $R^2 = 0.17$), so the predictive forecast (0.449) and especially the representational closure (0.754, table 4(a)) clear it; at the impact frame, by contrast, the same parameter floor on wake enstrophy reaches $R^2 \approx 0.70$, because the just-released gust nearly determines the instantaneous state and the gap opens only as the wake evolves. On the lift and the negative circulation, which track the gust forcing more directly, the floor matches or exceeds the predictive forecast, the same mixed ordering discussed in the text; out of distribution (test_c, $|G| = 4$) it turns negative for the wake, drag, and positive-circulation observables.

Observable	n_{train}	train R^2	test_b R^2	test_c R^2	JEPA d=64 test_b R^2
C_L	226	0.922	0.707	0.741	0.723
C_D	226	0.886	0.521	-3.993	0.634
I_y	226	0.651	0.417	0.207	0.056
wake-enstrophy	226	0.571	0.173	-0.103	0.449
circ. pos.	226	0.845	0.166	-1.884	0.201
circ. neg.	226	0.832	0.795	0.214	0.607

three independent retrains of both families the paired representational improvement is positive in all three (median +38, the predictive latent carrying the smaller error on 25 to 30 of 42 encounters), and the representational wake R^2 is 0.72 ± 0.10 for the predictive latent against -0.22 ± 0.14 for the reconstructive (mean ± s.d. over seeds). All six observables are reported in both modes, not a selected subset. The ordering is not uniform: the predictive latent leads on the wake observables (enstrophy and negative circulation), and this representational wake advantage is the one that survives both case-level clustering and a family-wide correction over the twelve paired tests; the forecast advantage points the same way but survives neither (table 10), and on forward forecast the scalar C_L and positive circulation fall within the paired noise between the families. The wake is where the three families separate. The reconstructive *representational* wake-enstrophy R^2 interval is entirely negative ($[-4.62, -0.12]$), a cleaner statement that the reconstructive latent fails to carry wake structure on held-out cases, and the confirmation rests on the mechanistic evidence of §4.3 (the drift ratio, the topological generator-count separation, robust over a defensible noise-floor and sampling range, the transport alignment, and the scale-resolved wake-enstrophy tracking), where the families separate consistently.

The conditioning-only floor (table 5) confirms that the wake closure is not an artefact of the parametric conditioning every model shares. Mapping the gust parameters $\mathbf{c} = (G, D, Y)$ to each observable at the forecast horizon $H = 16$ with a cross-validated kernel-ridge regressor, frame-matched to the closure it must be compared against, the floor on wake enstrophy is low (test_b $R^2 = 0.17$, rising to 0.31 only when a within-encounter shedding-phase proxy is added), well below both the predictive representational closure (0.754) and its forecast (0.449). The contrast with the impact frame is the mechanism: there the same parameter floor on wake enstrophy reaches $R^2 \approx 0.70$, because the just-released gust nearly determines the instantaneous state, but sixteen frames into the leading-edge-vortex roll-up the wake observable is governed by the unfolding dynamics, which is the information the predictive latent supplies and the parameters do not. The floor is by construction strong on the observables that track the gust forcing directly, the lift and the negative circulation, where it matches or exceeds the predictive forecast: the predictive advantage is specific to the spatially distributed wake, not uniform across observables. The wake-forecast claim

Table 6. Objective × architecture controls (decisive experiment). Held-out wake-enstrophy closure and mean closure (R^2 at $H = 16$, test_b, Markov rollout, mean ± standard deviation over three encoder seeds) for the four cells crossing {predictive, reconstructive} with {CNN, CNN+ViT} at $d = 64$, with the auxiliary heads (the instantaneous lift plus the 80-dimensional wake head) matched across all four cells, plus a predictive variant with the wake head removed. Predictive beats reconstructive on wake closure at both architectures (predictive versus reconstructive at CNN+ViT, and again at CNN), so the objective is not a confound of the architecture; removing the wake head from the predictive model collapses wake closure below the floor, so the wake supervision is also necessary. The CNN and CNN+ViT columns do not separate, so the ViT is not the driver.

Objective	Encoder	Aux. heads	wake-enst. R^2	mean R^2
predictive	CNN+ViT	lift + wake	$\mathbf{0.46 \pm 0.03}$	0.45
predictive	CNN	lift + wake	0.45 ± 0.06	**0.52**
reconstructive	CNN+ViT	lift + wake	0.16 ± 0.27	0.23
reconstructive	CNN	lift + wake	0.29 ± 0.05	0.42
predictive	CNN+ViT	lift only	-1.03 ± 0.29	0.09

therefore rests on the per-encounter paired test (table 10) and the drift mechanism (§4.3), and the conditioning floor establishes that the shared parameters do not supply the wake closure at the horizon the closure is evaluated. The converse control, a predictive model from which the conditioning is removed so it never sees **c**, is reported with the other ablations in §4.2 (table 7): it leaves the wake forecast at the floor as well, so the wake closure requires the latent and the parameters together, not either alone.

4.2. *Objective, architecture, and auxiliary-head controls*

The closure result of §4.1 is established, but the predictive latent *family* as configured closes the forward observables better than the reconstructive and linear families for reasons we have not yet separated: the predictive encoder differs from the reconstructive baseline not only in its training objective but also in its CNN+ViT architecture and its 80-dimensional wake supervision. Because the architecture and the wake supervision are confounds, we isolate the cause here, before turning to why the rollout works, its horizon dependence and drift mechanism (§4.3). The wake supervision is the more pointed confound, since the largest part of the reported advantage is on the wake observables that the predictive encoder alone was supervised on. Isolating the objective requires the controls in table 6: a 2×2 crossing of objective {predictive, reconstructive} with architecture {CNN, CNN+ViT} at matched $d = 64$, with the auxiliary heads matched across all four cells, evaluated under the same predictor and probes; together with a predictive variant from which the wake head is removed. The decisive question is whether the predictive objective still closes the wake observables better than the reconstructive one once the architecture and the wake supervision are held in common.

The controls (table 6, three encoder seeds per cell) give a two-part answer, and we state both. First, the predictive objective improves wake closure at *both* architectures with the auxiliary heads matched: the held-out wake-enstrophy R^2 is 0.46 ± 0.03 for the predictive CNN+ViT against 0.16 ± 0.27 for the reconstructive CNN+ViT, and 0.45 ± 0.06 for the predictive CNN against 0.29 ± 0.05 for the reconstructive CNN. The objective, not the architecture, carries the wake advantage: the CNN and CNN+ViT columns do not separate (the predictive CNN even leads on the mean-over-observables R^2, 0.52 against 0.45), so the ViT is not the driver. Second, and equally important, the wake-observable supervision is necessary: the predictive model with the wake head removed collapses to a wake-enstrophy R^2 of -1.03 ± 0.29, far below the floor and below every reconstructive cell. The wake-closure result therefore requires the predictive objective *and* the wake supervision together; it is not a property of the objective in isolation, nor of the architecture, nor of the wake head attached to an arbitrary objective (the reconstructive cells carry the same wake head and do not reach the predictive closure). We claim, accordingly, that the predictive objective improves forward wake closure over the reconstructive objective at matched architecture and matched supervision, while being explicit that the auxiliary wake head is a necessary component of the configuration that achieves it. The large seed variance of the reconstructive CNN+ViT cell (± 0.27) is itself consistent with the rollout-drift mechanism (§4.3): without a forward-predictable latent geometry, wake closure is unstable across initialisations.

Beyond the decisive crossing, three further controls vary the latent dimension, the second auxiliary head, and the predictor conditioning (table 7). Sweeping the predictive latent over $d \in \{4, 8, 16, 32, 64\}$ at fixed recipe, the held-out forecast holds on a plateau down to $d = 16$, with mean closure R^2 of 0.45, 0.36, and 0.36 at $d = 64$, 32, and 16, and then falls off: at $d = 8$ the wake-enstrophy forecast drops below the predict-the-mean floor ($R^2 = -0.07$) and the mean over observables halves to 0.17, and $d = 4$ is lower still (0.14). The reported $d = 32$ and $d = 64$ thus sit on the plateau of this curve, and the wake forecast is not recoverable below roughly $d = 16$ on this dataset. The instantaneous-lift head, removed while the wake head is kept (last row, the complement of the wake-removal control above), is load-bearing but less so than the wake head: at $d = 64$ its removal lowers the lift forecast R^2 from 0.72 to 0.54 and, through the shared latent dynamics, the wake forecast from 0.45 to 0.31 and the mean from 0.45 to 0.28, so the lift supervision improves even the channel it does not directly supervise. The two head ablations are therefore asymmetric: removing the wake head collapses the wake forecast below the floor (table 6), whereas removing the lift head degrades every channel without collapsing any. The third control removes the gust-parameter conditioning from the predictor (last row), so the model never sees $\mathbf{c}$. It leaves the representation almost intact (wake MAE 34.21 against 29.83, since the encoder is unconditional in either case) but degrades the forward forecast: the held-out wake-enstrophy R^2 falls from 0.449 to 0.038 and the lift from 0.723 to 0.392, with the mean at 0.309. The effect is stronger out of distribution: on the $|G| = 4$ test_c set the conditioned model holds a positive wake forecast ($R^2 = 0.33$) and a strong lift (0.79) while the unconditioned one collapses to -1.06 on the wake and to 0.39 on the lift, so the parameters are necessary to extrapolate the wake to an unseen gust strength. This is the exact complement of the conditioning-only floor of §4.1: the latent without the parameters and the parameters without the latent each leave the wake forecast near the predict-the-mean floor, and only their combination closes it.

4.3. *Why the rollout works: horizon, drift, topology, and transport*

The controls attribute the wake advantage to the predictive objective trained with wake supervision, but they leave open why a predictive latent rolls out stably where a reconstructive one does not. That explanation is geometric, and it shows first in how closure decays with the forecast horizon and then in three coordinate-free diagnostics of the rollout itself.

A dynamics paper should not rest on a single horizon. Comparing recursive (Markov) rollout against a one-shot prediction at horizon H, the two agree to within the encounter-level scatter for $H \leq 16$ and diverge for $H \geq 32$ as the recursive error compounds, which places the reported $H = 16$ inside the regime where the rollout mode does not affect the conclusion and which matches the planning horizon relevant to the impact-instant load. The full decay of closure with horizon, swept over $H \in \{1, 4, 8, 16, 32, 64\}$ for every family on test_b and test_c (figure 5), separates the families by the manner of their failure, not only its degree. The predictive latent degrades gracefully: its held-out wake-enstrophy forecast R^2

Table 7. Latent-dimension sweep and the two head/conditioning ablations for the predictive (JEPA) family, at $H = 16$ on the $n = 42$ test_b encounters, under the matched predictor and probes of §4.2. "Repr. wake MAE" is the representational mean absolute error of the wake enstrophy from the simulation-encoded latent (smaller is better); the R^2 columns are the forecast coefficient of determination of the Markov rollout (larger is better, the predict-the-mean floor is 0); "mean R^2" averages the six observables. The "wake only" row removes the instantaneous-lift head at matched $d = 64$ (the complement of the wake-removal control in table 6); the "no **c**" row instead removes the gust-parameter conditioning from the predictor at matched $d = 64$, so the model never sees (G, D, Y).

d	Configuration	Repr. wake MAE	C_L R^2	wake R^2	mean R^2
64	lift + wake	29.83	0.723	0.449	0.445
32	lift + wake	37.99	0.751	0.214	0.362
16	lift + wake	50.09	0.642	0.265	0.363
8	lift + wake	45.61	0.467	-0.072	0.174
4	lift + wake	58.51	0.181	0.194	0.137
64	wake only	35.44	0.542	0.313	0.276
64	lift + wake, no **c**	34.21	0.392	0.038	0.309

Table 8. Latent-drift diagnostic on test_b at $H = 16$. The Mahalanobis distance of the rolled-out latent is reported alongside that of the simulation-encoded latent in the same reference distribution; their ratio $r = d_{\text{Mahal, Markov}}/d_{\text{Mahal, DNS}}$ measures how far the autoregressive rollout has left the training manifold. A ratio near one means the rollout stays in distribution.

Baseline	d	$d_{\text{Mahal, ref}}$	$d_{\text{Mahal, rollout}}$	ratio r
Fukami	64	2.92	28.96	**9.90**
JEPA	64	8.08	6.84	0.85
JEPA	32	5.43	4.65	0.86
POD	64	7.74	6.27	0.81

stays above 0.5 to $H = 16$ at both $d = 64$ and $d = 32$, then declines. The reconstructive and linear latents fail abruptly: their wake-enstrophy forecast R^2 is already below 0.5 at $H = 1$ and negative by $H = 16$, so they never hold a usable wake forecast. The force coefficient C_L, which all families carry, decays smoothly for all of them. The abrupt wake failure coincides with the rollout-drift onset established next: the reconstructive latent leaves its manifold within a few steps, so its wake forecast collapses immediately rather than eroding.

The horizon behaviour points to a geometric cause, which the rest of this subsection establishes three ways that do not depend on the latent coordinates: a distance that shows whether the rollout leaves the training manifold, a topology that shows whether the encounter stays a single cycle, and an optimal-transport metric that shows whether latent distance tracks the physical rearrangement of the flow. Together they test the one property a predictive objective confers and a reconstruction objective does not, that the latent metric stays aligned with the transport geometry under the predictor step, so the rollout remains close to the encoded manifold.

First, the latent-drift diagnostic (table 8) measures how far each rollout leaves its training manifold. After $H = 16$ steps the reconstructive rollout sits an order of magnitude further from its training distribution than its simulation-encoded counterpart (ratio 9.90), while the predictive and linear rollouts remain within it (ratios 0.85 and 0.81). This is the mechanism behind the closure gap: the probe attached to a drifted rollout is queried outside the support it was fit on, so its output degrades for reasons unrelated to the predictor's one-step quality. A reconstruction objective places latent codes wherever the decoder finds convenient

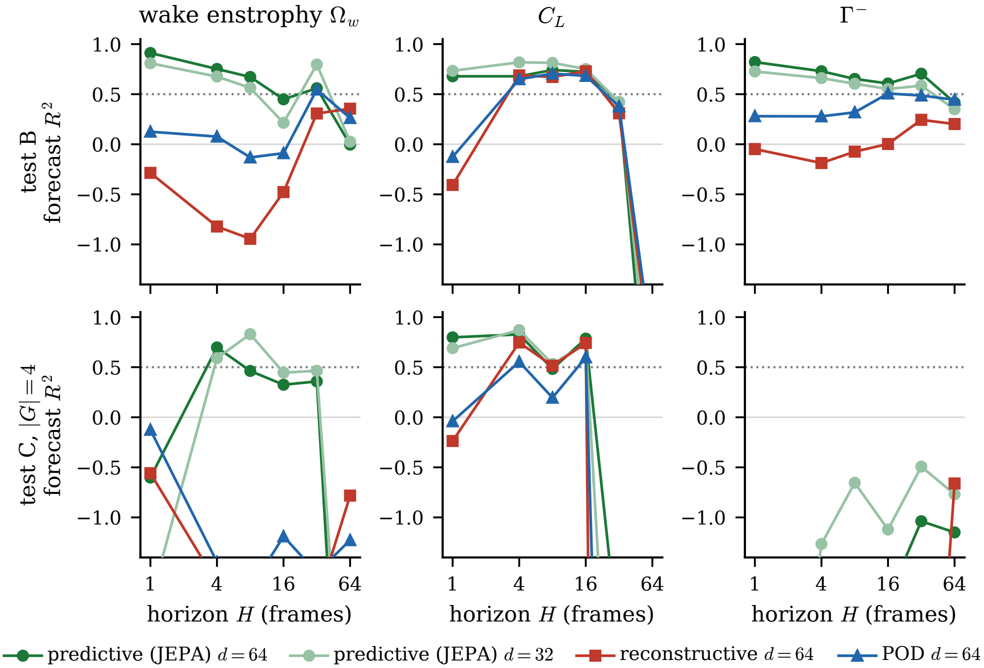

Figure 5. Held-out forecast closure (R^2 of the Markov rollout) versus horizon H for the wake-enstrophy, C_L, and negative-circulation observables, on the in-distribution test_b (top) and the $|G|$ = 4 extrapolation test_c (bottom), for the predictive latent at d = 64 and d = 32, the reconstructive autoencoder, and POD. The dotted line marks R^2 = 0.5. The predictive latent holds wake-enstrophy above the floor to H = 16; the reconstructive and linear latents are already below it at H = 1.

and does not constrain the geometry to be predictable in time; the predictive objective, regularised against collapse, produces a latent that supports stable rollout by construction. The pre-impact inference of the impact lift makes the same point from the encoder side: an oracle probe on the simulation-encoded predictive latent recovers the impact C_L at R^2 = 0.683 at every lead time, whereas the same oracle on the reconstructive latent is negative (−0.185), a representation failure rather than a probe failure, since no probe can recover information the latent does not carry. Rolling the predictive latent forward through the predictor recovers about 0.60 of that skill from five frames before impact and about 0.35 from ten, several frames of usable forecast horizon beyond a direct wall-pressure read at the same lead times.

Distance alone understates the geometric content of the result, so we next examine the trajectory topology. A gust encounter departs the baseline limit cycle, executes the LEV-formation and shedding excursion, and relaxes back toward it without closing within the window (§4.4); in the encoded latent the excursion nonetheless has the topology of a single cycle (Smith *et al.* 2024). We characterise this with persistent homology, which identifies the persistent one-dimensional cycle (H_1 generator) of a trajectory by a Vietoris–Rips filtration (Smith *et al.* 2024), applied here to four latent point clouds per encounter: the simulation-encoded predictive latent, its rollout, the simulation-encoded reconstructive latent, and its rollout (figure 6). The coordinate-free signature is not the survival of a single generator's lifetime, which is confounded by scale (the reconstructive rollout drifts into a large diffuse cloud that still supports a long-lived but non-canonical loop, so the rollout-to-encoded lifetime ratio is near one for both families and does not separate them).

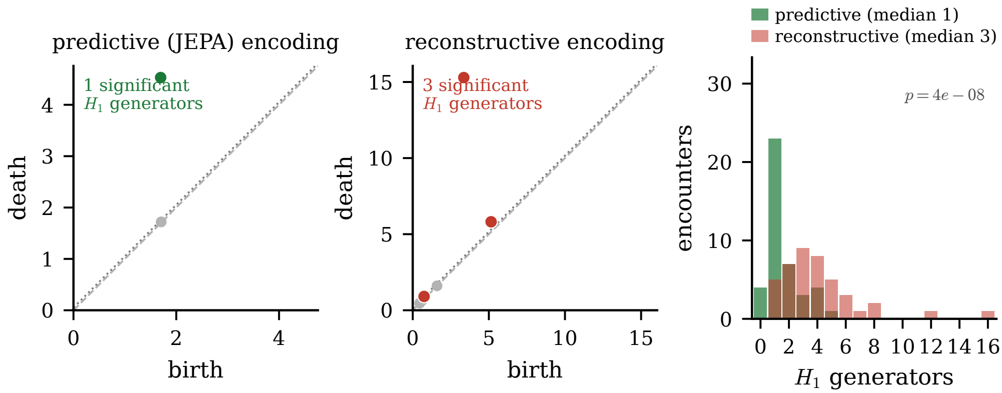

Figure 6. Persistent homology of the latent encounter on test_b. Left and centre: representative Vietoris–Rips H_1 persistence diagrams of the simulation-encoded predictive and reconstructive latents; coloured points are significant generators (lifetime above the dotted noise floor) and grey points are noise. Right: the distribution of the significant-H_1 generator count over the 42 held-out encounters, the scale-free signal. The predictive encoding is a single clean loop (median one generator) while the reconstructive encoding fragments (median three to four; Mann–Whitney one-sided $p = 4.4 \times 10^{-8}$).

It is the *generator count* of the simulation-encoded latent itself: the predictive encoding produces a clean single cycle (median one significant H_1 generator on test_b, 55% of encounters with exactly one), while the reconstructive encoding fragments into several (median 3.5, 71% with three or more; Mann–Whitney $p = 4.4 \times 10^{-8}$, and the same direction on test_c, $p = 0.04$). The predictive latent has the clean limit-cycle topology of the physical encounter; the reconstructive latent does not. This is consistent with the curvature geometry of §4.4: the impact frame is a curvature minimum of the predictive trajectory, a smooth pass-through rather than a sharp corner, so the cycle it traces is a single well-formed loop. The result at $d = 32$ is the same (single predictive loop versus reconstructive fragmentation). The separation is robust to the two free choices of the analysis, following the convergence protocol of Smith *et al.* (2024): across noise floors from 2% to 15% of the cloud diameter and trajectory subsampling down to 60 points the predictive median stays at one generator and the Mann–Whitney separation holds at $p < 10^{-3}$, and it survives case-level clustering (10 of 10 cases, signed-rank $p = 10^{-3}$ at the canonical setting). At the most aggressive 20% floor, or under heavy subsampling to 30 points, the spurious reconstructive generators are themselves filtered and the two medians converge, so the exact p is floor- and sampling-dependent while the median-count separation is not (Appendix A).

The third test makes the geometry physical by comparing latent distances with the unbalanced optimal-transport distance between vorticity fields. Euclidean distance between vorticity fields is insensitive to the spatial transport of structures, whereas an unbalanced optimal-transport (OT) distance measures the cost of rearranging one field into another and so tracks advection of the LEV and shear layer (Tran *et al.* 2026). Computing the OT distance between simulation fields along an encounter gives a physical transport geometry against which the latent metric can be checked: the prediction is that distances in the predictive latent track the OT distances more faithfully than distances in the reconstructive latent, so that iterating the predictor step keeps the rollout closer to the data manifold. The prediction holds: the per-encounter Spearman correlation between the latent distance matrix and the simulation OT-distance matrix on test_b is 0.63 for the predictive latent at $d = 64$ (0.61 at $d = 32$) against 0.45 for the reconstructive latent, a margin of 0.18, with the predictive latent more faithful on 36 of 42 encounters. The predictive latent metric is therefore more aligned with the physical transport geometry, in this empirical distance-matrix sense, consistent with a rollout that remains closer to the encoded manifold. We report this per encounter rather than pooled across encounters: pooling conflates the within-encounter geometry with a between-encounter latent-norm scale that the drift-prone reconstructive latent inflates, which reverses the pooled statistic and is itself a symptom of the same drift.

4.4. *Where closure succeeds and fails in physical space*

The mechanism explains why the predictive rollout stays on the manifold; we now ask where in the gust envelope, and where in the flow field, that closure holds. The parameter dependence reveals one clear weakness. Probing the predictive latent for the gust parameters on test_b recovers the gust strength G and the core diameter D ($R^2 = 0.46$ and 0.80) but barely resolves the wall-normal offset Y ($R^2 = 0.10$). This is a property of the data rather than of the architecture: the training mass concentrates near $Y \approx 0$, so encounters at different offsets within the envelope are weakly separated along Y, and the latent inherits that weakness. Any deployment-time parameter estimate should treat Y as a nuisance variable the encoder resolves only weakly.

The same latent can also be read relative to the undisturbed shedding cycle. Because the undisturbed wake is a limit cycle (§2.1), an encounter is naturally described by its phase and amplitude relative to that cycle, the same reduction used to design gust-mitigation control for these flows (Fukami *et al.* 2024). Reading the predictive latent this way makes "recovery" precise: it is the return of the latent trajectory to the baseline limit cycle. The predictive latent of the undisturbed case does trace a closed periodic orbit (figure 7): concatenating the four no-gust episodes, the return distance is 1.0% of the orbit diameter, and the four episodes overlap the same loop, with a shedding period of about 56 frames (2.8 convective times, Strouhal number near 0.36). A phase advances monotonically along it (via the Hilbert analytic signal of the two leading latent principal components), though the orbit lives in more than two dimensions, so the phase is a reduction rather than a complete coordinate. A gust encounter departs this orbit and is carried back toward it under the predictive rollout while the reconstructive rollout departs further: the median return-to-orbit distance on test_b is smaller for the predictive latent at every horizon, and the separation is bootstrap-robust at $H = 64$ (predictive 8.70 versus reconstructive 9.96, 95% confidence interval on the paired difference $[1.13, 3.25]$), marginal at $H = 32$. One caveat applies: the simulation gust trajectories do not themselves fully return to the baseline orbit within the 120-frame window. That window is the gust-release period itself ($6\,t/c$ at $\Delta t = 0.05\,t/c$, with impact near $2\,t/c$), so only about $4\,t/c$ of post-impact relaxation is ever observed before the next gust is released, short of a full return; this is a window-length limitation set by the gust-release cadence, and at fixed core diameter it is not strength-dependent. The comparison therefore measures the relative drift direction (the predictive rollout contracts toward the orbit, the reconstructive one expands away), not a completed physical recovery. The result is the same at $d = 32$. This limit cycle and the persistent cycle of the previous subsection are the same object viewed dynamically and topologically.

We can also locate where in the envelope the predictive advantage concentrates. Rather than a per-stratum closure table, which the present test set is too small to populate, we read the per-encounter paired improvement $\Delta e = e_{\mathrm{AE}} - e_{\mathrm{JEPA}}$ in held-out wake-enstrophy error directly against each gust axis with a locally weighted regression and a bootstrapped band (figure 8). The advantage is not uniform across the envelope. It grows with the gust core diameter D (Spearman $\rho = 0.42$ on the encoded latent, 0.53 under forecast) and toward off-midplane wall-normal offset Y ($\rho = 0.49$ and 0.45), both significant, while its

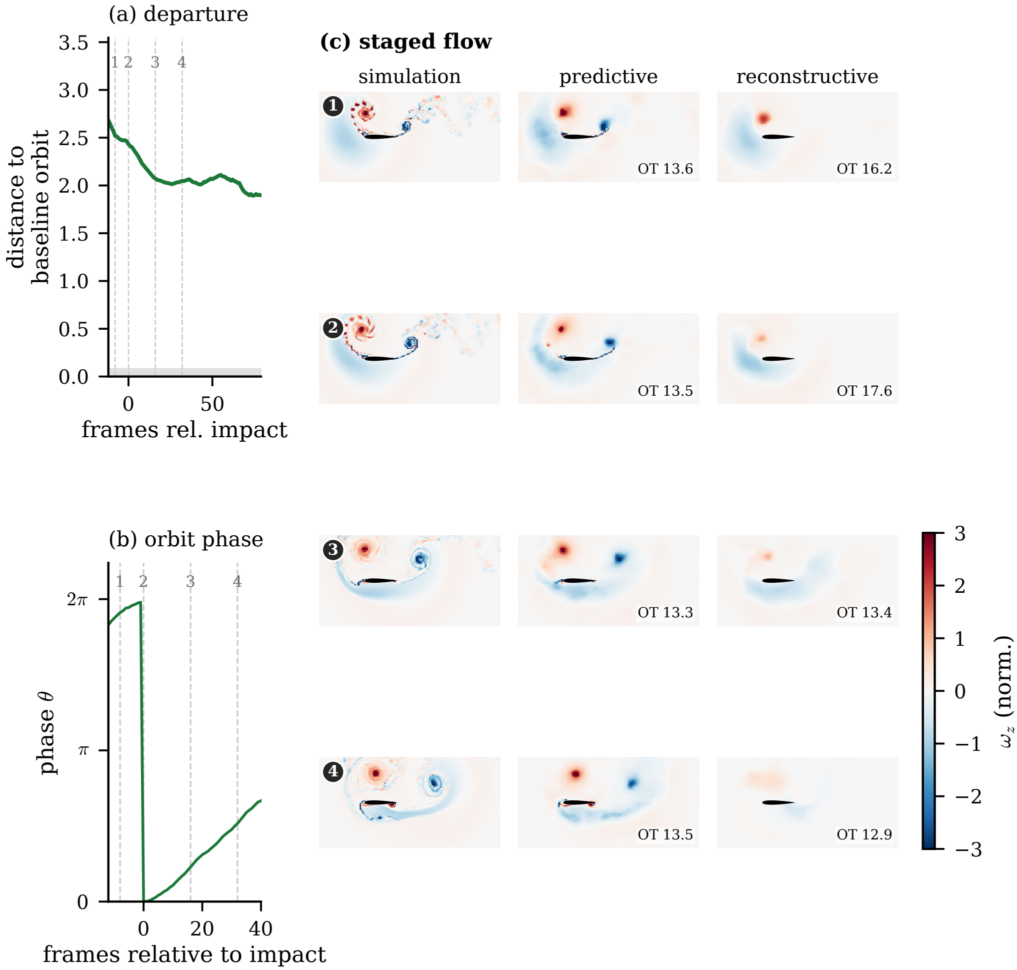

Figure 7. The gust encounter in the predictive latent, for a strong negative gust $(G, D, Y) = (-3.0, 1.5, -0.1)$. (a) Distance of the gust trajectory to the baseline orbit (full latent, in units of the orbit diameter; the grey band is the baseline orbit thickness): the encounter departs the baseline cycle and only partially relaxes back, without closing within the inter-gust window; the single-cycle topology of the encoded latent is established coordinate-free in figure 6. (b) The orbit phase relative to the impact frame. (c) Mid-plane vorticity at the four stages (rows) for the simulation, the predictive decode, and the reconstructive decode (columns); the predictive decode localises the impingement while the reconstructive decode collapses, with the optimal-transport field distance between each decode and the simulation annotated at each stage (largest for the reconstructive decode near impact). The numbered stages are those of figure 4.

dependence on the gust strength G is weak on the encoded latent and moderate under forecast. The predictive advantage therefore concentrates away from the $Y \approx 0$ training mass and at the larger cores that most reorganise the wake, consistent with the weak Y resolution of the probe above. The shedding phase at impact, the axis the gust-response literature emphasises (Fukami *et al.* 2024), is only weakly sampled by this dataset: impact is fixed near a single convective instant while the gust recurs about every two shedding periods, so the encounters fall into essentially two phase clusters and a phase-resolved trend is not identifiable here, which a design that varies the impact phase would isolate.

These parameter and phase readings are statements about numbers; the encounter is itself a statement about vortices. Decoded and reconstructed vorticity fields on held-out

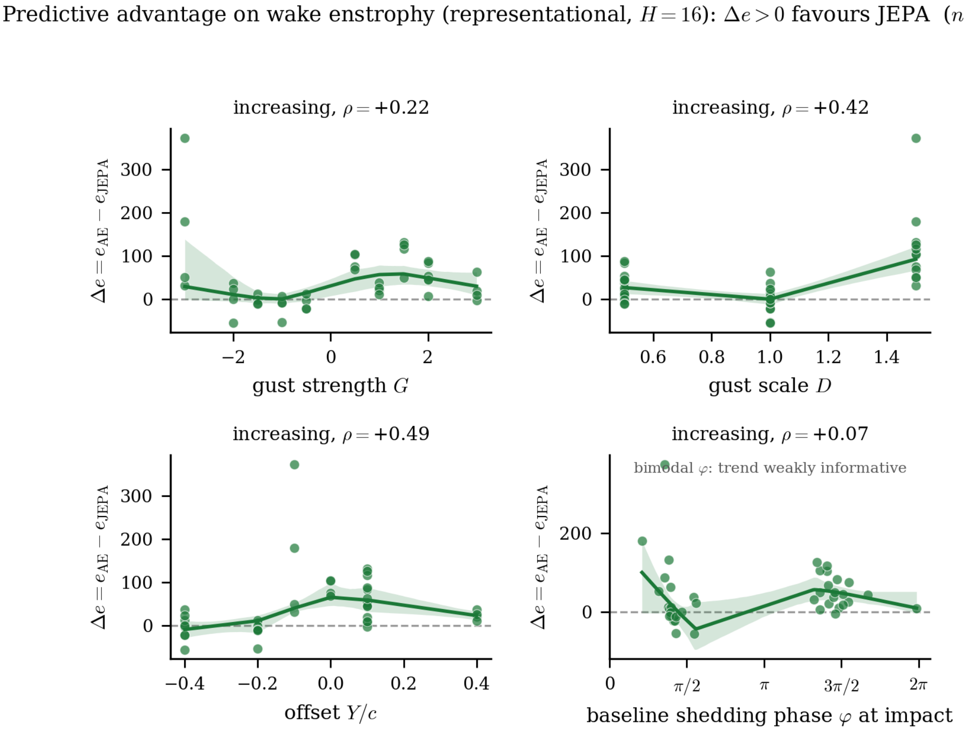

Figure 8. Per-encounter paired improvement $\Delta e = e_{\mathrm{AE}} - e_{\mathrm{JEPA}}$ in held-out wake-enstrophy error against the gust strength G, core diameter D, wall-normal offset Y, and the baseline shedding phase at impact ϕ (test_b, $n = 42$); positive Δe means the predictive latent carries the smaller error. Each panel shows the per-encounter values, a locally weighted (LOWESS) trend, and a bootstrap band over encounters. The predictive advantage grows with D and toward off-midplane Y; the shedding-phase axis falls into two clusters because impact is fixed near a single convective instant.

encounters (figure 9) connect the two, with one methodological caution and one new metric. Two scoping points apply throughout this subsection. The fields are encode-decode reconstructions, the visualisation decoder applied to the simulation-encoded latent, so the quantitative comparisons report the decoder's ceiling on what each latent carries from a faithfully encoded field, not a forecast rollout, whose latent leaves the data manifold for the reconstructive baseline (§4.3). And every quantitative physical-space claim is made on the validated large-scale band ($\sigma/c = 0.05$); the visualisation decoder is blurry at small scale by construction, so the large-scale leading-edge vortex and shear layer are tracked meaningfully while full-resolution and small-scale field fidelity are not claimed.

The caution is that field-space similarity scores are misleading here. On a held-out interpolation encounter the reconstructive autoencoder attains a high structural similarity yet a peak reconstructed vorticity magnitude of only 0.36 against a target peak near 9.7: it has collapsed to a near-uniform mean field and lost the vortex-impingement signature, and the high similarity merely reflects bulk agreement with a field that is itself mostly quiescent away from the impact region. The predictive latent localises the impingement at the right position and sign (peak magnitude 5.31) but its visualisation decoder is blurry by construction, and the linear basis is amplitude-accurate (peak 4.10) since its objective is amplitude-optimal.

The new metric resolves the caution. An unbalanced optimal-transport distance between the reconstructed and the simulation fields, computed for the positive and negative vorticity

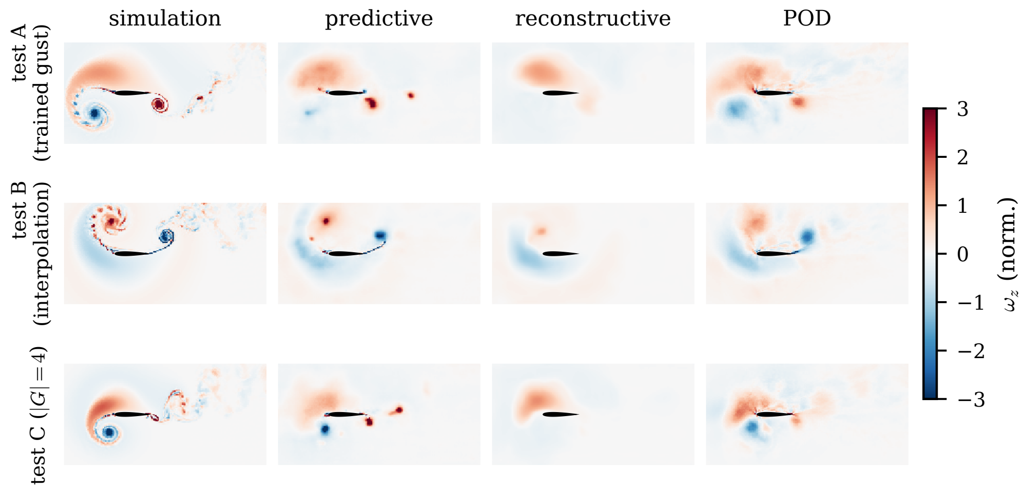

Figure 9. Decoded vorticity at the impact frame on held-out encounters from the three splits (rows: a trained-gust test_a encounter, an interpolation test_b encounter, and a $|G| = 4$ test_c encounter) for the simulation and the predictive, reconstructive, and POD decodes (columns). The predictive decoder is blurry by design but localises the impingement; the reconstructive autoencoder collapses toward a near-uniform field on the held-out interpolation condition, while the linear basis is amplitude-accurate. All families degrade in the $|G| = 4$ regime, reported as the observability boundary of §2.1 rather than as a generalisation claim. Field-space similarity is reframed by the optimal-transport metric introduced below.

separately and summed (Tran *et al.* 2026) (entropic unbalanced Sinkhorn transport cost with a KL marginal relaxation, fields pooled to 48×24 for tractability), penalises the collapsed reconstruction correctly and reorders the comparison where structural similarity misleads (figure 10). At the impact frame on test_b the transport distance is largest for the reconstructive autoencoder (11.25) and smaller and comparable for the predictive decode (9.90) and the linear basis (9.95). The instructive case is the linear basis: it carries the *highest* structural similarity of the three (0.69 against 0.65 for the predictive decode), which would wrongly crown the energy-optimal reconstruction as the best, yet under transport it is no better than the blurry predictive decode and far better than the collapsed autoencoder, which is the physically correct ordering. We therefore lead with the transport distance and report similarity alongside for continuity. On the OOD test_c set ($G = +4$) the transport ordering breaks (the reconstructive distance falls below the predictive one), so the transport advantage is an in-envelope result, of a piece with the three-dimensional observability boundary that degrades every metric there.

Finally, the wake observables tie to the structures that carry the load. A Gaussian scale decomposition (large scale at filter width $\sigma/c = 0.05$, following (Odaka *et al.* 2026; Motoori & Goto 2019)) separates the large-scale vortical structures responsible for the lift peaks from the fine scale, applied to the simulation field and to each family's reconstruction (figure 11). On test_b the large-scale wake enstrophy of the predictive reconstruction tracks the simulation closely (Pearson correlation 0.89 at impact and 0.91 at $H = 16$, relative error 0.22), while the reconstructive autoencoder collapses it (correlation 0.23 and 0.61, relative error near 0.8, retaining only 16 to 20% of the large-scale amplitude and almost none of the small-scale energy): it smooths the leading-edge vortex and shear layer away. The linear basis has comparable large-scale correlation to the predictive latent but worse amplitude (relative error 0.34 to 0.38), so the claim is specific: the predictive latent tracks the simulation's large-scale wake structure better than the reconstructive autoencoder on both correlation and amplitude, and better than the linear basis on amplitude. Across the

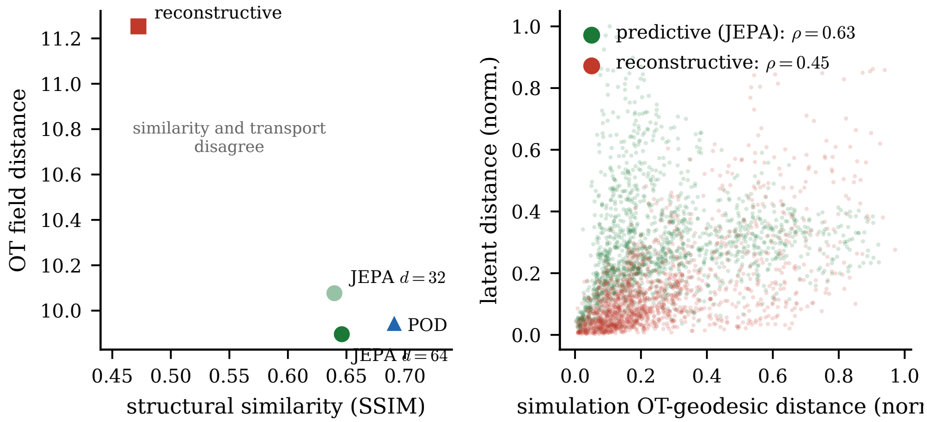

Figure 10. Optimal-transport reconstruction comparison (Tran et al., JFM 2026) on test_b. Unbalanced OT field distance by family at the impact frame, with the structural-similarity ranking shown alongside; the linear basis has the highest similarity but no transport advantage over the predictive decode, while the collapsed reconstructive field is correctly penalised. The Shepard panel shows the per-encounter alignment of latent distance with the simulation OT distance, predictive versus reconstructive.

staged encounter the simulation large-scale enstrophy rises from the gust-induced leading-edge flux through the LEV roll-up and peak-load shedding; the predictive reconstruction follows that rise while the reconstructive one stays flat. This is where the wake-enstrophy advantage of §4.1 becomes a statement about the leading-edge vortex and shear layer rather than about a scalar. Tracking the large-scale leading-edge vortex directly makes this quantitative. Extracting the dominant suction-side vortex from the same $\sigma/c = 0.05$ field, the predictive decode places a coherent leading-edge vortex in all 42 held-out encounters at $H = 16$, within 0.32 chords of the simulation centroid and recovering its circulation to within 5%, whereas the reconstructive decode has smoothed the vortex below detection in 36 of the 42 (its large-scale leading-edge amplitude falls to about a tenth of the simulation), and where it survives its circulation error is six times larger. Per encounter the wake-enstrophy error and the leading-edge-vortex circulation error co-vary (Spearman 0.57), so the scalar wake observable stands in for the position and strength of the leading-edge vortex that produces the lift transient. The result holds at $d = 32$; on the OOD test_c set the predictive tracking degrades (0.91 to 0.65 from impact to $H = 16$), as expected where the flow becomes three-dimensional and the mid-plane decomposition is itself incomplete (§2.1).

4.5. *Wall-pressure observability*

At deployment the vorticity field is not available; only the wall pressure is. The forecast result has an exact deployment mirror: we ask which representation is most recoverable from a sparse set of K wall-pressure taps over a pre-impact window, using one kernel-ridge estimator across families for a uniform comparison (the sensor placement, the lead-time prediction, and a closed-loop pilot are detailed in Appendix B). The predictive state is markedly the most recoverable (figure 12a): with $K = 8$ taps the held-out test_b state R^2 is 0.89 at $d = 64$ and 0.87 at $d = 32$, against 0.58 for the reconstructive autoencoder and 0.32 for the linear basis at $d = 64$, and the contrast is sharpest at matched capacity. The energy-optimal POD coordinates, optimal for reconstruction, are the hardest of the three to read from the wall: the same predictive objective that produces a forecastable wake state produces one that is also legible from pressure.

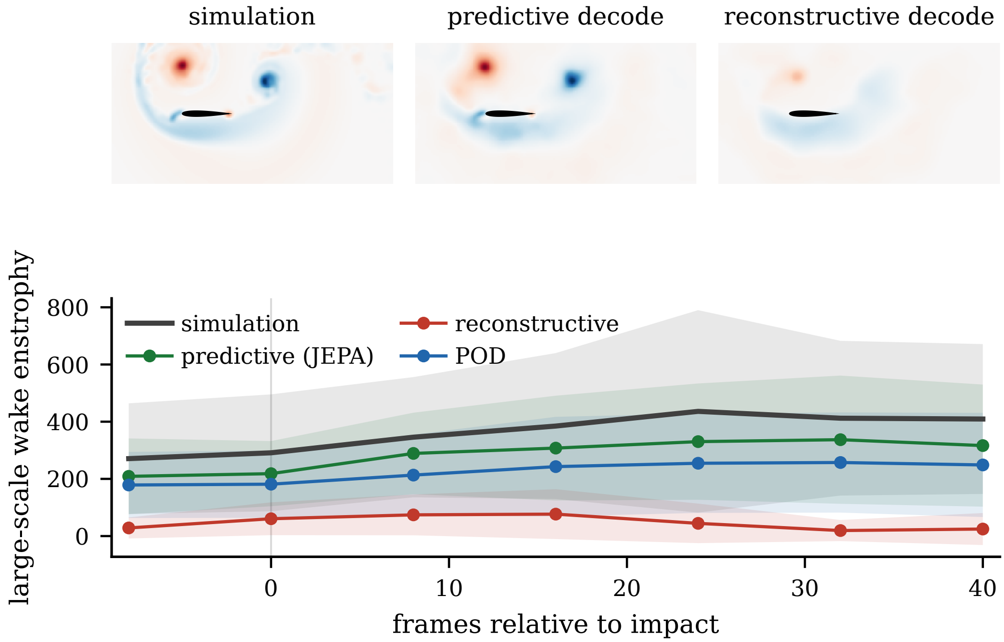

Figure 11. Gaussian scale decomposition ($\sigma/c = 0.05$) of the held-out reconstructions. Top: large-scale vorticity for the simulation, the predictive decode, and the reconstructive autoencoder at $H = 16$; the predictive latent retains the leading-edge vortex and shear layer the reconstructive one smooths away. Bottom: large-scale wake enstrophy through the staged encounter for the simulation, predictive, and reconstructive fields, on each split.

Recoverability is not estimation quality, and the distinction is itself informative (figure 12b). The three-dimensional reconstructive latent is the *easiest* to recover from pressure (state $R^2 = 0.91$ at $K = 2$, the highest of any family) yet routing the impact-lift estimate through it gives the *worst* lift error (mean absolute error 1.0 against a C_L spread of 1.37), while the predictive latent gives 0.49; the lift itself is read most accurately straight off the pressure (0.39), because the imminent lift is already imprinted on the wall. The contribution of the predictive latent is therefore the forecastable, observable state it hands to the predictor, not the instantaneous lift, which is read more accurately straight from the wall.

5. Discussion

5.1. *Interpretation and scope*

The central finding is geometric, and it concerns what the latent represents as much as how it evolves. Under a shared predictor, parametric conditioning, and probe family, the predictive latent carries the wake structure the reconstructive and linear families do not, an advantage concentrated on the spatially distributed wake observables and established first at the representation level, where it survives case-level clustering and a family-wide correction; the forward rollout is consistent with the same ordering but is not, on its own, decisive at the case level. The mechanism is a property of the latent metric: it is, to a good approximation, aligned with the optimal-transport geometry of the flow in the empirical distance-matrix sense, so that the rollout stays close to the data manifold. A reconstruction objective fixes the latent only up to a diffeomorphism (Tran *et al.* 2026; Kelshaw & Magri 2024), so the decoder is free to place codes anywhere it finds convenient

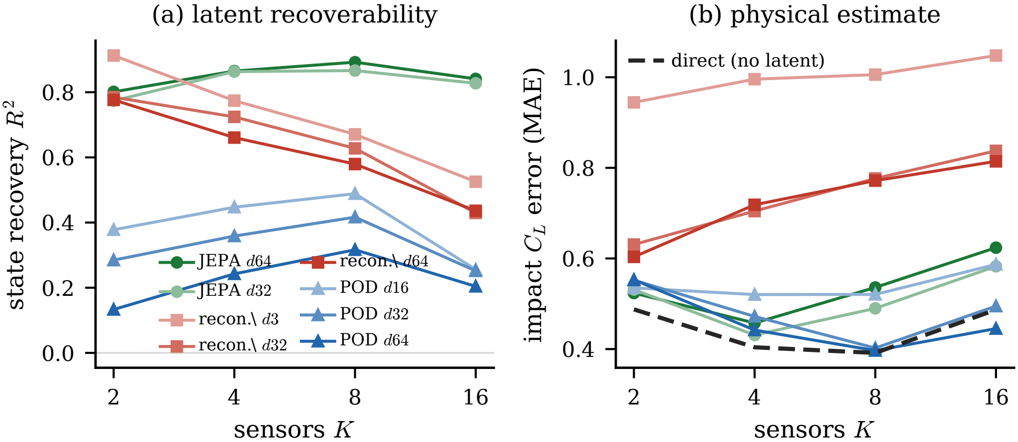

Figure 12. Cross-family pressure observability under the target-conditioned (TCSI) sensor placement. (a) Latent-state recovery R^2 versus sensor count K: the predictive (JEPA) state is the most recoverable at $K \geq 4$, the linear basis the least. (b) The physical estimate, impact C_L mean absolute error routed through each family's recovered latent, with the direct pressure-to-C_L baseline (dashed). The reconstructive $d = 3$ latent is the easiest to recover (highest R^2 at low K) yet gives the worst lift estimate, and the lift is read most accurately directly from the pressure.

and nothing constrains the latent geometry to be predictable in time; under autoregressive rollout the reconstructive latent therefore leaves its own training manifold, and a probe queried on that drifted state fails for reasons unrelated to the predictor's one-step accuracy. The predictive objective, regularised against collapse, instead organises the latent so that the encounter is a recurrent cycle that survives rollout, the on-manifold property that data-driven state-space and Koopman models of coherent-state dynamics on invariant manifolds rely on (Constante-Amores & Graham 2024). The manifold departure of the reconstructive rollout is the parametric-gust counterpart of the catastrophic-divergence tail reported for reconstruction-based latent dynamics in controlled wakes (Solera-Rico *et al.* 2025); the predictive objective is the common cure. The wake observables are the sharpest discriminator precisely because they can change while the scalar forces do not: a representation can carry the lift signature of an encounter without carrying the vorticity redistribution that produces it, and the reconstructive latent does exactly that, recovering the impact lift poorly from its own oracle while the predictive latent recovers it well.

A coordinate-level reading of the three latents makes the mechanism concrete and explains why the wake, and not the forces, separates the families. For each family we fit a linear probe of the future wake enstrophy on the $d = 64$ latent and ask how that forecast is distributed across the coordinates. The predictive latent forecasts the future wake at a held-out rank correlation of 0.83 that no single coordinate approaches (the best one reaches 0.44), so the information is a *collective* code spread across many coordinates. Grouped by what they encode, these coordinates fall into a large wake-vorticity group (about fifty of the sixty-four, tracking the wake circulation and thickness) and a smaller gust-forcing group (about eleven, tracking the gust ratio and the forces), the remainder being near-silent; within each group the coordinates are strongly correlated rather than a disentangled basis, and on their own the wake-vorticity coordinates forecast the future wake at about 0.7 and the forcing coordinates at 0.45, neither matching the full latent's 0.83, so the forecast draws on both groups together. The reconstructive and linear latents forecast it far less well (0.60 and 0.56) and from no collective structure: their best single coordinate is already as informative as the whole latent, a near-zero gap that holds across three reconstructive-encoder seeds

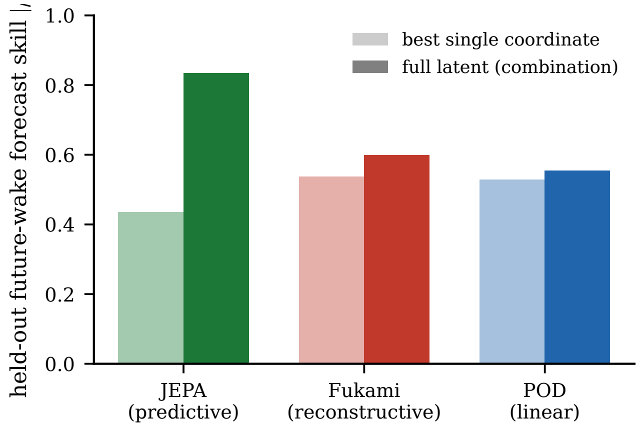

Figure 13. The predictive objective builds a collective wake-forecast code that the reconstructive and linear objectives do not. For each encoder family at matched $d = 64$, the held-out forecast skill of the future wake enstrophy $\Omega_w(t+H)$ at $H = 16$ from the best single latent coordinate (light) and from the full latent combination (dark), on test_b. The predictive (JEPA) latent shows a large gap (combination 0.83, best single 0.44): the wake forecast is distributed across many coordinates. The reconstructive (Fukami) and linear (POD) latents show a near-zero gap and lower skill, so their wake forecast is no better than a single coordinate. Observational probes on the frozen encoders.

(figure 13). The forces, by contrast, are forecast redundantly in every family, with dozens of individual coordinates carrying them. An energy or reconstruction objective therefore retains the force signature, which is present in many coordinates, but not the distributed combination that forecasts the wake, which is exactly the observable on which the predictive latent leads. The predictive forecast direction also anticipates the wake beyond the gust parameters and the instantaneous forces (held-out rank-partial 0.83), so it carries genuine forecast information and not merely the present forces. These are observational probes on the frozen encoders, a latent-space reading of the closure result of §4.1 rather than a separate claim, and they sit alongside the information-theoretic modal analyses of related flows (Martínez-Sánchez *et al.* 2023; Arranz & Lozano-Durán 2024; Koshikawa *et al.* 2026).

This organisation survives a halving of the latent dimension, which is the basis for reading it as a low-dimensional predictive state rather than an artefact of capacity. The latent is in fact far lower-dimensional than its nominal budget: the encoded $d = 64$ latent has a participation ratio of about 1.7 across the held-out trajectories, with the leading principal component carrying roughly three-quarters of the variance and six components spanning 90%. The dominant direction tracks the gust-strength dynamic range, the largest physical source of variation across encounters, so the structure is concentrated in a handful of directions rather than collapsed, as the parameter probes and forecasts confirm; both $d = 64$ and $d = 32$ are therefore over-complete relative to it, and the halving leaves the representation and every mechanism diagnostic intact. At $d = 32$ the predictive latent matches the $d = 64$ latent on representation quality, the held-out wake-enstrophy representation R^2 is 0.74 against 0.75, and on every mechanism diagnostic: it traces the same single persistent cycle rather than fragmenting (§4.3), its latent metric aligns with the physical transport geometry to within the $d = 64$ margin (rank correlation 0.61 against 0.63), it returns to the baseline limit cycle under rollout (§4.4), it retains the same large-scale wake structure (§4.4), and its rollout-drift ratio is 0.86 against 0.85 (table 8). The one quantity that degrades is the in-distribution wake forecast, where the $H = 16$ closure R^2 falls from 0.45 to 0.21 (table 4(b)): halving the dimension costs forecast sharpness on the wake observables while leaving the geometric mechanism intact. Halving once more to $d = 16$ keeps the qualitative ordering, the predictive latent still leads on wake enstrophy where the reconstructive and linear latents are negative (table 4), but its representational wake R^2 now falls to 0.55 (from 0.74 at $d = 32$): the budget is approaching the effective dimensionality, and the extra dimensions buy precision rather than a qualitatively different representation. On the out-of-distribution test_c set the $d = 32$ latent is in fact marginally the more robust of the two, 0.45 against 0.33 on wake enstrophy. The physics the predictive objective organises into the latent, the closed encounter cycle and its transport-aligned metric, is therefore a low-dimensional object already present at $d = 32$, and the additional dimensions at $d = 64$ buy in-distribution forecast precision rather than a qualitatively different representation. The parameter content of the two latents is consistent with this reading: the dominant gust-strength axis is encoded as well at $d = 32$ as at $d = 64$, while the core diameter and the wall-normal offset, the axes the data resolve least, are the ones whose recovery degrades when the dimension is halved, the same axes whose forward closure the extra dimensions sharpen.

The mechanism above invites a strong reading, so we are deliberate about the scope of the claim. The predictive encoder differs from the reconstructive baseline in three respects at once: the training objective, the CNN+ViT architecture, and the wake supervision. The conditioning-only floor removes the shared parametric conditioning as an explanation, and the matched-architecture, matched-auxiliary 2×2 controls of §4.2 separate the remaining two. They give a two-part answer that we state in full. The predictive objective does improve forward wake closure over the reconstructive objective at both a CNN and a CNN+ViT encoder with the auxiliary heads matched, and the two architecture columns do not separate, so the gain is the objective and not the architecture. The wake supervision, however, is a necessary part of the configuration that achieves it: a predictive model with the wake head removed collapses below the predict-the-mean floor on wake closure. The claim is therefore that the predictive objective improves forward closure *when trained with wake supervision*, not that the objective in isolation does, and not that the wake head attached to an arbitrary objective does, since the reconstructive cells carry the same head and do not reach the predictive closure. The linear basis, for its part, remains genuinely useful: it is the most competitive family on the integrated impulse and is amplitude-accurate in reconstruction, so the result is not that POD is obsolete but that it does not produce a latent whose rollout closes the wake observables.

5.2. *Limitations and outlook*

Three limitations bound the claims. First, the wall-normal offset Y is weakly resolved because the training mass concentrates near $Y \approx 0$; this is a data limitation, and Y-conditional statements should be read with it in mind. Second, the $|G| = 4$ extrapolation degrades for a physical reason, not merely a statistical one. As set out in §2.1, the encoder observes a single mid-plane vorticity slice whose information no longer determines the three-dimensional $|G| = 4$ dynamics (Fukami *et al.* 2025), so test_c is the observability boundary of that representation, not a parametric-generalisation claim. Third, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss; a reader expecting sharp reconstructions from a predictive latent will be disappointed, which is the explicit trade the objective makes. To these we add, as a fourth, the co-necessity of the wake supervision established in §5.1: the predictive objective improves forward closure in the configuration tested, with the wake head as a necessary component, not as a property of the objective stripped of all auxiliary supervision.

These limitations come into sharpest focus against the use the architecture is ultimately meant for. The predictor is, by construction, the latent dynamics function a model-based controller would plan against, and three properties of the architecture are relevant to that use: closure holds out to the planning horizon relevant to the impact-instant load, the latent is two to three orders of magnitude cheaper to evolve than the full field, and the rollout stays on the training manifold, which is the property that determines whether long-horizon plans remain meaningful since a planner queries the predictor on states reached by its own rollout. Because the encoder is unconditional, the latent itself is the observable a controller has access to, but the conditioning **c** is not observed at deployment and must be estimated from the wall or marginalised over before the predictor is run, and the gust parameters enter only the predictor; replacing the parametric conditioning with an actuation channel is a change of input rather than of architecture, and the phase-amplitude reading of §4.4 connects the predictor directly to the sensitivity-function control designed for these flows (Fukami *et al.* 2024). An actuation channel would enter at the same point as the gust, the predictor conditioning, so the same forward-closure machinery would apply with the control as the action; consistent with treating the world-model view of §1 as motivation only, we caution against over-reading this as an interventional world model. The conditioning is far from inert: removing it entirely collapses the held-out wake forecast to the predict-the-mean floor (§4.2), so the gust parameters carry forecast-relevant information, and what is at issue is the narrower question of whether the predictor responds to a *change* in that conditioning as the simulation does. A direct test, perturbing the conditioning along G by a fixed increment and comparing the predicted change in the six closure observables to the change measured between matched simulation encounters that differ only in G, does not support the interventional reading: across the parameter grid the predicted and measured responses are uncorrelated to weakly anti-correlated (pooled $r = -0.27$ at the best-powered increment $\Delta G = 2$, $n = 23$ base points), although the predictor's response has physically plausible magnitude. The predictor is therefore a conditional forward model that forecasts the observed encounters well and whose conditioning improves closure, not a validated counterfactual model of parameter changes; it is not itself a controller, and the pilot below shows a further gap.

As a limitation rather than a result, a closed-loop pilot that chains a wall-pressure state estimator, the predictor rollout, and an observable probe does not yet meet the C_L and I_y tolerances set in advance; the bottleneck is the rollout rather than the estimator, since the within-tolerance fraction is essentially unchanged when an oracle latent replaces the estimated one, so the representation and the pressure observability are in place but the rollout needs reinforcement before a usable controller follows (estimator, sensor-placement, and loop details in Appendix B).

The contribution is narrow by design: a controlled comparison on a single parametric flow, anchored to forward physical closure rather than to in-distribution reconstruction, with a drift mechanism that explains the gap and a set of controls (§4.2) that bound its interpretation. What the comparison establishes is bounded accordingly. The predictive advantage is robust on the wake observables and in the drift mechanism; it is partial on parameter extrapolation, where the wake forecast survives at $G = +4$ but the other observables do not (§4.1), and on the wall-normal offset Y, which is harder to resolve than G and D (§4.4); and it is absent where we tested the interventional reading and the closed-loop tolerances, both of which fail. We claim the closure result and its mechanism, not a controller or a counterfactual model. The natural next step is to apply the same matched-predictor closure protocol to other parametric flow datasets, to test how general the predictive-versus-reconstructive trade reported here actually is.

6. Conclusions

We asked what makes a reduced-order state forecast parametric vortex-gust airfoil interactions, and reached a geometric answer: the state forecasts well when its latent metric stays well aligned with the optimal-transport geometry of the flow, in the empirical sense that latent distances preserve the ordering of the transport distances, so that the rollout stays close to the data manifold. A reconstruction objective does not impose this property and its rollout drifts off the manifold; a predictive objective regularised against collapse does. We tested this by comparing a predictive latent (JEPA), a reconstructive observable-augmented autoencoder, and a proper orthogonal decomposition basis as reduced states at Re = 5000, under a protocol in which a single autoregressive predictor, conditioning, and probe family were held fixed so that only the encoder varied, judging each by forward physical closure rather than by reconstruction. On held-out cases the predictive latent carries the wake structure the others do not: its representational wake-enstrophy closure is positive where the reconstructive and linear baselines are negative, a factor of two to three closer to the simulation on wake enstrophy, and the per-encounter improvement survives case-level clustering and a family-wide correction, while the linear basis remains most competitive on the integrated impulse. The same separation appears coordinate-free in the rollout drift, the encounter topology, the optimal-transport geometry, and the large-scale leading-edge vortex. The forward forecast points the same way and is consistent with this, though at the case level it is not on its own decisive. Because the predictive encoder also differs in architecture and auxiliary supervision, matched controls attribute the advantage to the predictive objective trained with wake supervision. The same predictive state is the most recoverable of the three from sparse wall pressure, so it is observable as well as forecastable, the property a downstream estimator or controller would exploit. We claim a conditional forward-closure model, not a validated counterfactual controller: the predictor forecasts the consequences of observed gust parameters but is not shown to respond correctly to interventions on them. The remaining limitations are the weak resolution of the wall-normal offset, the three-dimensional observability boundary at $|G| = 4$, and a rollout error that still prevents a closed-loop control demonstration.

Funding. *[Authors to insert funding sources and grant numbers.]*

Declaration of interests. The authors report no conflict of interest.

Data availability statement. The reduced-order-model and evaluation code, the configuration files, the v2 data-split manifest, and the analysis outputs that regenerate the tables and figures are deposited at https://doi.org/10.xxxx/zenodo.PLACEHOLDER under the MIT License (DOI to be finalised on acceptance). The processed per-encounter cache, or a representative subset sufficient to rerun the heavier steps, is available from the corresponding author. The raw direct numerical simulations were computed with the SOD2D solver (Gasparino *et al.* 2024) and are owned by the simulation collaborators.

Author contributions. *[Authors to insert contributions, for example in CRediT format.]*

Use of generative-AI tools. Generative-AI assistance (Anthropic Claude, accessed through the Claude Code command-line interface) was used during 2026 to help with data analysis, figure generation, and manuscript drafting. All numerical results were computed by the authors' own code and verified against the underlying data, and the authors take full responsibility for the content of the article.

Appendix A. Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail

The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that is 288 spatial tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the

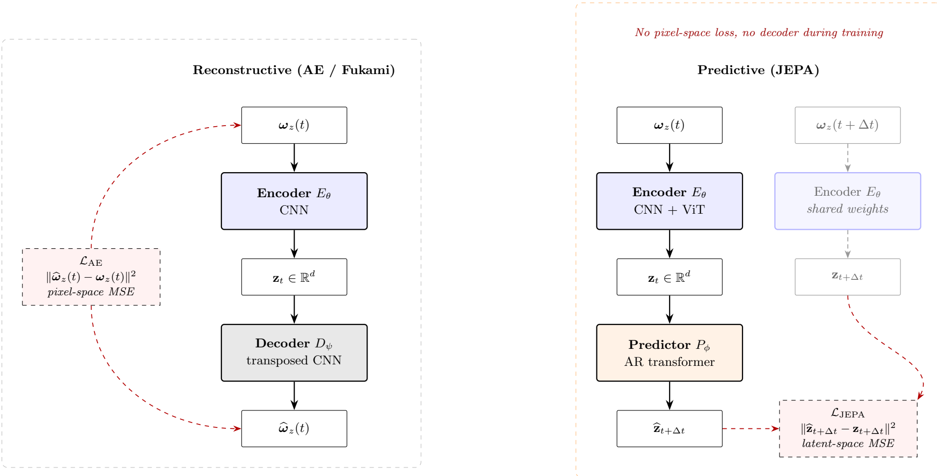

Figure 14. Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t, decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training. This is the contrast the matched-predictor protocol of §3.2 evaluates; the visualisation decoder is trained only afterwards, on the frozen encoder.

latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The predictor is a six-layer autoregressive transformer of width 384 with sixteen heads and dropout 0.1, with a causal mask and rotary temporal positions. The gust parameters $\mathbf{c} = (G, D, Y)$ enter only the predictor, through adaptive layer normalisation initialised to the identity (AdaLN-Zero), so that at initialisation the predictor is the unconditioned dynamics and the conditioning is learned as a modulation. Training minimises the one-step latent prediction loss plus a scheduled-sampling rollout term and the regulariser below, with AdamW and bf16 mixed precision.

Anti-collapse regularisation

Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line (Maes *et al.* 2026; Balestriero & LeCun 2025), with M random one-dimensional projections of the latent and a small weight (0.01) in the loss. As a safeguard the participation ratio of the latent is monitored every thousand iterations, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

Uncertainty quantification

Every held-out coefficient of determination and mean absolute error in the paper carries three independent uncertainty signals, following the reporting protocol fixed with the split. The first is a bootstrap confidence interval from 2000 resamples over the held-out test encounters, which is the dominant source of scatter at the present sample size ($n = 42$ for test_b, $n = 24$ for test_c) and is wide for the individual wake observables, as noted in §4.1. The second is the variance across three encoder retrains from independent random seeds, reported as the standard deviation in the objective-times-architecture controls of §4.2. The third is a five-fold cross-validation over cases on the read-out (probe) step, which guards against a probe that overfits the particular held-out split. Together these approximate what a full five-fold case-level cross-validation of the entire pipeline would buy, at a fraction of the training cost; a single split with bootstrap intervals is the community standard for this class of study.

Table 9. Training-set fit, for reference only. Coefficient of determination R^2 for predicting the six observables from the rolled-out latent ($H = 16$) evaluated *on the training distribution*. This quantifies in-distribution fit and is *not* the headline; held-out closure is in table 4. Best per column in bold; row mean at right.

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
JEPA	64	**0.864**	**0.802**	0.562	**0.934**	**0.922**	**0.927**	**0.835**
JEPA	32	0.863	0.795	0.490	0.898	0.901	0.897	0.808
Fukami	3	0.553	0.408	0.191	0.079	0.130	0.033	0.232
Fukami	32	0.695	0.596	0.278	0.333	0.369	0.313	0.430
Fukami	64	0.671	0.589	0.335	0.277	0.336	0.352	0.427
POD	16	0.763	0.720	0.514	0.574	0.582	0.527	0.613
POD	32	0.763	0.748	0.690	0.589	0.618	0.621	0.671
POD	64	0.641	0.656	**0.799**	0.373	0.388	0.506	0.561

Table 10. Paired held-out closure, predictive ($d = 64$) against reconstructive ($d = 64$) on the $n = 42$ test_b encounters at $H = 16$, for the representational mode (probe on the simulation-encoded latent) and the forecast mode (Markov rollout from impact). Δerr is the per-encounter mean of (reconstructive − predictive) absolute error in each observable's own units, so a positive value favours the predictive latent; the 95% interval is a 2000-resample paired bootstrap, and the sign test counts the encounters on which the predictive error is the smaller of the pair. The wake enstrophy is the pre-registered primary endpoint (§2.2); the final column is its raw sign-test p Holm-corrected across the twelve tests of the table, so the secondary observables are read after the family-wide correction. The 42 encounters come from only 10 cases, so the wake rows additionally carry a case-clustered 95% interval (resampling the 10 cases, encounters averaged within case): representational $[+12.5, +77.2]$ and forecast $[-4.5, +72.6]$, with a case-level signed-rank test on the per-case mean improvement of $p = 0.03$ (7/10 cases) and $p = 0.10$ (7/10 cases) respectively. The representational wake advantage survives both the case-level clustering and the family-wide correction; the forecast wake advantage survives neither. All six observables are shown, not a selected subset.

Mode	Observable	Δerr	95% CI	sign test	Holm p
repr.	C_L	+0.364	$[+0.070, +0.643]$	25/42, $p = 0.14$	0.98
repr.	C_D	+0.039	$[-0.052, +0.131]$	20/42, $p = 0.68$	1.00
repr.	I_y	+0.52	$[-0.028, +1.08]$	25/42, $p = 0.14$	0.98
repr.	wake-enst.	+43.1	$[+23.5, +66.0]$	31/42, $p = 1.4\times10^{-3}$	0.017
repr.	circ. pos.	+0.256	$[-0.117, +0.673]$	20/42, $p = 0.68$	1.00
repr.	circ. neg.	+1.19	$[+0.567, +1.93]$	30/42, $p = 4.0\times10^{-3}$	0.044
forecast	C_L	−0.021	$[-0.206, +0.152]$	20/42, $p = 0.68$	1.00
forecast	C_D	+0.066	$[+0.011, +0.128]$	26/42, $p = 0.082$	0.74
forecast	I_y	+0.264	$[-0.164, +0.666]$	26/42, $p = 0.082$	0.74
forecast	wake-enst.	+32.0	$[+10.8, +54.8]$	27/42, $p = 4.4\times10^{-2}$	0.44
forecast	circ. pos.	−0.018	$[-0.319, +0.297]$	19/42, $p = 0.78$	1.00
forecast	circ. neg.	+0.555	$[+0.067, +1.11]$	18/42, $p = 0.86$	1.00

Training-set fit

For reference, table 9 reports the in-distribution training fit that the held-out closure of §4.1 must be read against. The predictive latent attains the highest training fit (mean $R^2 = 0.835$ at $d = 64$), but the ordering of the families is established by the held-out closure of table 4, not by this in-distribution number.

Paired closure across all observables

Table 10 reports the per-encounter paired comparison of §4.1 for all six observables in both modes, the source of the wake-enstrophy significance cited in the main text. Pairing cancels the encounter-to-encounter difficulty that widens the marginal intervals of table 4.

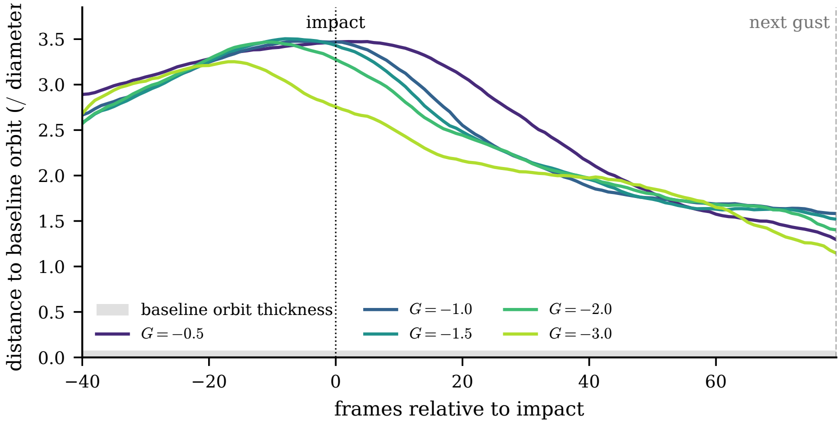

Figure 15. Diameter-controlled return to the baseline orbit. Distance of the encoded gust trajectory to the no-gust orbit (predictive latent, in orbit diameters; grey band is the orbit thickness), at fixed core diameter $D = 1.0$ and $G < 0$, averaged over each case's encounters, for five gust strengths. The departure amplitude is nearly independent of $|G|$ and all strengths are still contracting when the next gust is released at +79 frames, so the non-return is a window-length effect, not a strength-dependent one.

Topological robustness

The significant-H_1 generator count of §4.3 declares a generator significant when its persistence exceeds a noise floor set at a fraction of the cloud diameter (the largest finite H_0 death), the canonical analysis using 5% of that diameter and all 120 trajectory frames. Following the convergence protocol of Smith *et al.* (2024) we sweep the floor over $\{2, 5, 10, 15, 20\}$% and the points per encounter over $\{120, 60, 40, 30\}$ (uniform subsampling), 20 settings in all. The predictive median is one generator in every setting. The predictive-versus-reconstructive Mann–Whitney separation is below 10^{-3} for floors from 2% to 15% at full (120-point) sampling and for floors up to 5% at half (60-point) sampling (the canonical 5%, 120-point value is 4.4×10^{-8}), while the reconstructive median falls from six generators at a 2% floor to one at a 20% floor as the aggressive floor filters the spurious generators. Under heavy subsampling to 30 points both medians approach one and the separation washes out. The robust statement is therefore the median-count separation over a defensible floor and sampling range, not the exact p; the case-clustered signed-rank test (10 of 10 cases at the canonical setting) holds across the same range. The full grid is in the released analysis package.

Return to the baseline orbit within the encounter window

The load-bearing caveat of §4.3, that the simulation gust trajectories do not fully return to the baseline orbit within the 120-frame encounter, is a consequence of the gust-release cadence rather than a strength-dependent failure to recover. Figure 15 holds the core diameter fixed at $D = 1.0$ with $G < 0$, averages the distance to the baseline orbit over each case's encounters, and sweeps $|G|$. The departure amplitude near impact is nearly independent of gust strength (peak distance 3.2 to 3.5 orbit diameters across $|G| \in [0.5, 3.0]$), every trajectory is still contracting toward the orbit at the end of the window, and the strongest gust ends closest rather than farthest. The residual offset at the window's end is therefore set by the $\sim 4\,t/c$ of post-impact relaxation the cadence allows and by the core diameter, a larger core leaving a more persistent wake, not by $|G|$. This window-truncated return is common to all three encoders and does not affect the relative drift comparison of §4.3, which contrasts the predictive and reconstructive rollouts over the same window.

Appendix B. Sparse wall-pressure state estimation

At deployment the vorticity field is not available; only the wall pressure is. We ask how much of the latent state, the gust parameters, and the impact lift can be recovered from a sparse set of K wall-pressure taps over a pre-impact window, and through which representation. The input is the span-averaged pressure on the 192 surface taps, and the pressure-to-latent map is a kernel-ridge regressor with a radial-basis kernel on the flattened K-sensor by pre-impact-window feature vector. Nonlinear estimators of this class (kernel ridge, a temporal-convolutional network, a regularised perceptron, and a cross-validated LSTM) perform comparably on the state; the recurrent model's advantage appears for the scalar lift at short lead (figure 18). We use kernel ridge throughout the cross-family comparison for a uniform, low-variance estimator across families. This is a feasibility study, not a closed controller, and we report where it works and where it does not.

Target-aware placement helps only when sensors are scarce

The placement we use, TCSI, is *target-aware*: a greedy forward selection that, for each of the 192 surface taps, forms a pre-impact pressure window of $W = 17$ frames centred on impact and adds at each step the tap whose inclusion most improves the recovery of the target, the leading principal component of the predictive latent, from the joint flattened window of the already-selected taps (the exact selection objective is in the deposited code). The picks are nested in K and cluster on both surfaces near the leading edge, where the gust impinges (figure 16a). We compare it against the QR-pivot optimum of the pressure modes (qDEIM) (Manohar *et al.* 2018), which is *target-blind*: it places taps to reconstruct the pressure field, with no knowledge of the latent it is later asked to recover. The two differ in what they optimise, so the comparison measures the value of target-awareness, not of placement quality alone, and that value is confined to the scarce-sensor regime. At $K = 2$ the target-aware placement recovers the predictive state at $R^2 = 0.80$ against 0.44 for the target-blind one, but by $K \geq 4$ the wall pressure is information-rich enough that the two converge ($R^2 \approx 0.86$ to 0.89; figure 16b). The cross-family recoverability result of §4.5 is therefore not an artefact of a bespoke target-aware placement: with four or more taps a placement blind to the target recovers the predictive latent just as well. The target-aware selection is greedy and joint rather than a marginal-attribution ranking (SHAP or permutation importance): under the strongly collinear surface pressures, scoring taps individually double-counts the shared information, whereas conditioning each pick on those already chosen spreads the array. A direct comparison against log-likelihood and mutual-information selectors of the same target, all of comparable cost, confirms that the greedy selection gives the best held-out latent recovery and that the alternatives are no better.

Cross-family recoverability (in the main text)

The headline cross-family result, that the predictive state is the most recoverable from the wall and that recoverability is not estimation quality, is reported in §4.5 (figure 12). Decoding the pressure-estimated latent makes that recovery physical (figure 17): from eight taps the leading-edge vortex and shear layer are recovered close to the oracle decode, and even two taps capture the gross impingement.

Parameters and the out-of-distribution boundary

The gust parameters recover directly from pressure on test_b, with the core diameter best and the wall-normal offset weakest ((G, D, Y) $R^2 = (0.73, 0.89, 0.65)$ at $K = 8$), in keeping with Y's weak resolution throughout the paper. On the out-of-distribution test_c set ($|G| = 4$) every estimator and every family gives negative state R^2 (between -0.3 and -0.9) and a large lift error, the same three-dimensional observability boundary of §5.2 that bounds the rest of the model.

Predicting the impact before it occurs

The deployment target is the impact itself, ahead of time. Ending the pre-impact pressure window τ frames before impact and predicting the impact-frame state and lift (figure 18), the impact state stays recoverable to eight instants ahead with a gentle degradation, and kernel ridge and the LSTM are comparable across the lead (state R^2 between 0.80 and 0.83). The impact lift is read accurately to about six instants ahead before degrading. Both estimators are selected by five-fold cross-validation on the training set, and the selection is not overfit: the cross-validated state R^2 (0.83) matches the held-out value (0.80), and the search prefers a small recurrent model. The LSTM reads the scalar lift more accurately at every lead (mean absolute error 0.25 at the impact frame and 0.39 at a six-frame lead, against the kernel ridge's 0.30 and 0.45; lift $R^2 = 0.89$ against 0.77 at the impact frame), an advantage that survives cross-validated selection and reflects its use of the temporal signature of the approaching gust. The imminent impact is therefore predictable a handful of instants before it occurs, the regime a controller would act in, and degrades into the rollout-limited regime of the closed-loop pilot beyond that lead.

The closed loop is rollout-limited, not estimator-limited

Chaining the pressure estimator, the predictor rollout, and the observable probe into a closed loop does not yet meet a tight control tolerance: at $H = 16$ only 18% of test_b encounters fall within 10% of the true C_L, against a target of 80%. The informative point is that this gate fails even for an oracle that is handed the true impact latent and the true parameters (18% as well), so the tolerance is bounded by the irreducible error of the predictor and probe, not by the pressure estimator. The honest reading is that the representation and the pressure observability are in place in distribution, and the closed loop is limited by the rollout horizon rather than by the estimator; more frequent re-estimation from pressure during the rollout, or a longer scheduled-sampling horizon, is the route to a usable controller. We report this as a pathway, not a result.

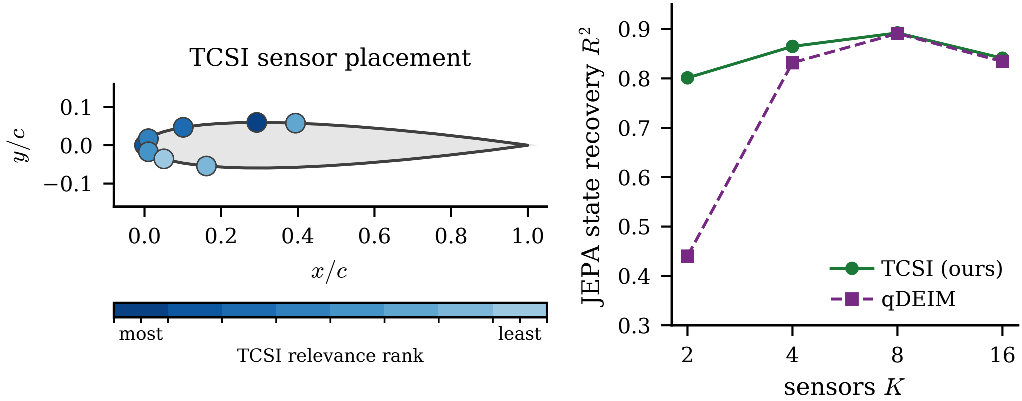

Figure 16. Sparse-sensor placement. (a) The target-conditioned (TCSI) taps on the NACA 0012 surface, a single ranked set with one marker per tap coloured from dark (picked first, most informative) to light (least); the greedy picks are nested, so the two darkest are the decisive $K = 2$ pair and the eight shown are the $K = 8$ set. They cluster near the leading edge, where the gust impinges. (b) Predictive-state recovery (held-out test_b R^2) versus sensor count for the target-aware selection (TCSI) and the target-blind QR-pivot optimum of the pressure modes (qDEIM): the target-aware placement helps at $K = 2$ but the two converge by $K \geq 4$, so the recoverability result is not an artefact of the placement.

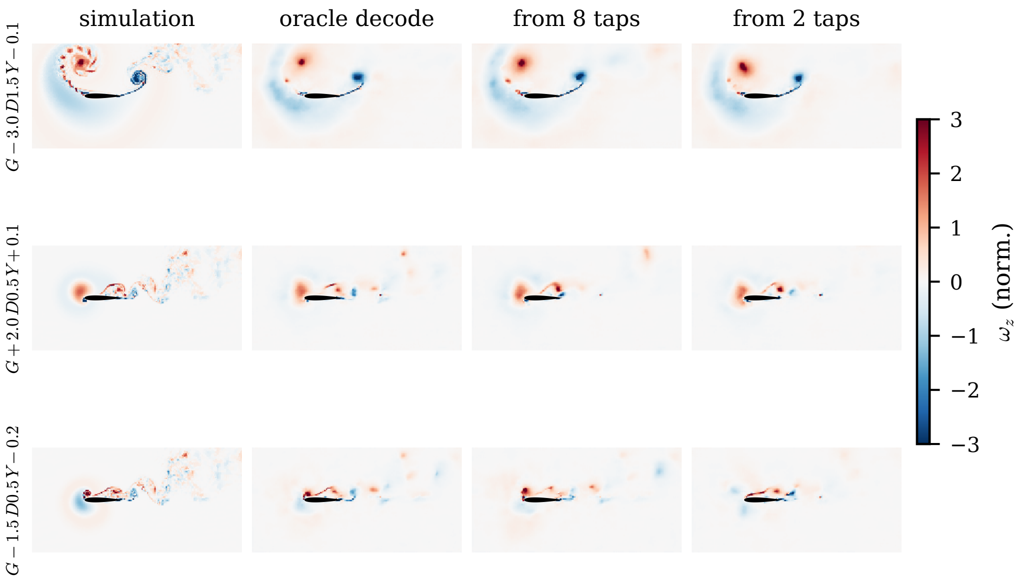

Figure 17. Flow recovered from sparse wall pressure on three held-out encounters (rows, labelled by (G, D, Y)). Columns: the simulation, the oracle decode (the visualisation decoder applied to the simulation-encoded latent), and the decode of the latent estimated from $K = 8$ and $K = 2$ TCSI pressure taps. From eight taps the predictive latent recovers the leading-edge vortex and shear layer close to the oracle; two taps capture the gross impingement. The decoder is blurry by design (§4.4).

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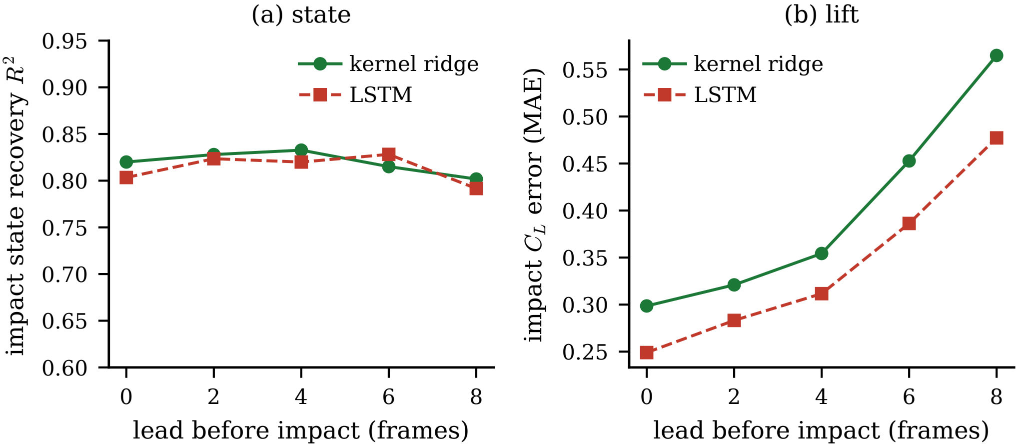

Figure 18. Predicting the impact before it occurs. (a) Impact-state recovery R^2 and (b) impact-lift C_L mean absolute error versus the lead, the number of frames before impact at which the pre-impact pressure window ends, for the kernel-ridge and LSTM estimators ($K = 8$ TCSI taps, test_b). The impact state is recoverable to eight instants ahead; the lift to about four, with the LSTM the more accurate lift estimator at short lead.

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